Auditing Fairness in Large Language Models
-
Detection and Mitigation Strategies

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Abstract

Large Language Models (LLMs) frequently absorb and magnify social stereotypes, creating fairness concerns. This study built a unified audit pipeline to spot and curb those biases, focusing on stereotype detection and mitigation. The pipeline assessed several recent open-source LLMs—representing different sizes and architectures—using intrinsic benchmarks (like CrowS-Pairs, StereoSet, and WEAT-style tests) and extrinsic tasks (including WinoBias and BBQ). Pervasive bias emerged across models, though its strength varied significantly depending on model characteristics and the specific benchmark used; notable gaps between intrinsic and extrinsic bias scores were observed. Models designed with a reasoning focus generally exhibited lower bias on demanding extrinsic tasks compared to more general-purpose models. Mitigation trials, focused on a model exhibiting significant baseline bias, then applied inference-time self-debiasing and parameter-efficient (LoRA) fine-tuning. While each technique individually yielded only modest reductions in stereotype rates on the BBQ benchmark, their combination proved highly synergistic, resulting in a substantial decrease in the tendency to select stereotype-aligned answers. These results confirm that systematic auditing is practical and demonstrate that pairing lightweight fine-tuning with inference controls offers an effective and resource-efficient pathway towards fairer model behaviour, although broader validation of generalisation and utility impacts remains essential future work. The findings also suggest that architectural choices favouring explicit reasoning may confer inherent fairness advantages.

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Chapter 1

Introduction & Motivation

As AI (Artificial Intelligence) systems are increasingly used in sensitive social contexts, bias and fairness in AI have emerged as critical issues. Large Language Models (LLMs) inherit and occasionally increase societal biases found in the massive volumes of human-generated text that they are trained on [1]. These models have demonstrated impressive capabilities in understanding and generating text, yet they may produce outputs that reflect stereotypical associations or unfair assumptions about people's gender, race, profession, or other protected characteristics (attributes such as age, disability, religion, sexual orientation, etc., for which discrimination is often legally or ethically prohibited). Such biased behaviour can lead to harmful outcomes, from offending users to reinforcing discrimination in critical applications.

For instance, risk assessment tools used in the U.S. justice system have shown racial bias, disproportionately flagging Black defendants as higher risk [2], while automated hiring tools have penalised female candidates based on biased historical data [3]. These real-world examples demonstrate the severe consequences, affecting liberty and opportunity, when algorithmic bias goes unchecked in domains like hiring or criminal justice. Therefore, a major difficulty in modern AI development is ensuring that LLMs treat all groups equitably – i.e., match them with ideals of justice and equality.

Motivation: The motivation for this project stems from the need to audit LLMs for bias and to develop techniques to mitigate any bias detected. Recent incidents and studies have illustrated that even state-of-the-art models can exhibit biases (e.g., associating certain occupations exclusively with one gender, or answering questions differently based on implied demographic traits of the subject) [4, 5]. These issues, coupled with the demonstrated real-world harms of algorithmic bias, motivate a systematic approach to measuring bias in LLM behaviour. By rigorously evaluating an LLM across a spectrum of bias tests, we gain insight into where and how the model may be unfair. Furthermore, exploring mitigation strategies addresses the follow-up question: once we know a model has a bias, what can be done about it?

From an ethical perspective, improving model fairness is consistent with the goals of responsible AI, which is to develop AI systems that do not cause harm or represent certain groups of people negatively. Thus, this project is motivated by the practical need to make AI systems more equitable as well as the scientific interest in understanding LLM biases.

Aim: In light of the above, the aim of this project is to create a thorough auditing framework for assessing the fairness of LLMs and to put bias mitigation techniques into practice, which will increase the model's fairness without significantly impacting its overall performance. In essence, the project seeks to both detect and reduce bias in LLMs in a systematic way.

Objectives: To achieve this aim, several concrete objectives were established:

1. **Literature Review:** Conduct an in-depth review of research on fairness in LLMs and their ethical risks. This involves reviewing existing evaluation metrics, bias issues in NLP applications, and

definitions of bias and fairness. *Measurable outcome:* A summarised background knowledge (Chapter 2) that informs the rest of the project.

2. **Fairness Definitions:** Using the literature review as a guide, develop precise working definitions of bias and fairness for LLM outputs. These definitions help scope what types of bias the project will focus on (for example, stereotypical bias in model predictions). They will be illustrated with examples to ensure practical clarity. *Measurable outcome:* a set of fairness criteria or metrics to be used in the auditing framework.

3. **Bias Auditing Framework Development:** Design and implement a bias auditing framework to assess the LLM. This framework should cover both intrinsic bias (bias in the internal representations of the model or language modeling behaviour) and extrinsic bias (bias that shows up in task-specific performance). In practical terms, this involves integrating several bias detection tests (metrics and benchmarks) into one toolkit. The implementation will be done using `Python`, utilizing datasets like `CrowS-Pairs`, `StereoSet`, and libraries like `Hugging Face Transformers` for model access. *Measurable outcome:* A functional software pipeline that can consistently prompt the model and measure different bias metrics.

4. **Mitigation Strategies Implementation:** Investigate and implement techniques to mitigate biases identified in the detection phase. Strategies such as parameter-efficient fine-tuning, specifically Low-Rank Adaptation (LoRA), and inference-time self-debiasing will be explored. Each strategy, individually and in combination, will be applied to a selected model (`Qwen2-1.5B`), and its effect on a chosen bias benchmark (`BBQ`) will be assessed using metrics derived from objective 3. *Measurable outcome:* a collection of experimental findings comparing the model's bias metrics prior to and following mitigation, allowing for the evaluation of the most successful tactics on the target benchmark.

Notably, as the project progressed, the scope was refined to focus on stereotype-related biases in English-language LLM outputs, rather than other fairness aspects such as calibration, which are outside the current scope. Throughout the project I handled sensitive bias-related data carefully. I ensured that any potential sensitive or offensive content in my results was reported in an academic, respectful way, complied with the university's ethics guidelines, and required no extra approval because I used only publicly available data and involved no human subjects.

Dissertation Structure The dissertation is arranged so that each chapter builds naturally on the last, guiding the reader from context to reflection.

- **Chapter 1 – Introduction** sketches the practical and ethical reasons for examining bias in contemporary language models and states the overarching aim that threads through the rest of the work.

- **Chapter 2 – Background Research** surveys the critical literature on algorithmic fairness, catalogues the most widely–used bias benchmarks, and considers current mitigation strategies, setting a clear reference point for the project's methodological choices.

- **Chapter 3 – Methodology** describes how the theoretical concepts were translated into practice for both the detection and mitigation phases. It explains the selection of datasets and models, the design of evaluations (intrinsic and extrinsic for detection; focused on BBQ for mitigation), the implementation of the mitigation techniques (self-debiasing, LoRA fine-tuning), and the overall experimental framework.

- **Chapter 4 – Results** presents the empirical findings in detail, first reporting the outcomes of the bias detection phase across all evaluated models and benchmarks. It then presents the results from the mitigation experiments conducted on Qwen2-1.5B using BBQ, discussing what these results

reveal about the effectiveness of the applied strategies and considering trade-offs between fairness improvement and potential performance impacts.

- **Chapter 5 – Conclusion** reflects on what has been learned from both the detection and mitigation phases, acknowledges the study's limitations, and suggests concrete directions for extending the work.

Figure 1.1 offers a visual roadmap of this research's workflow.

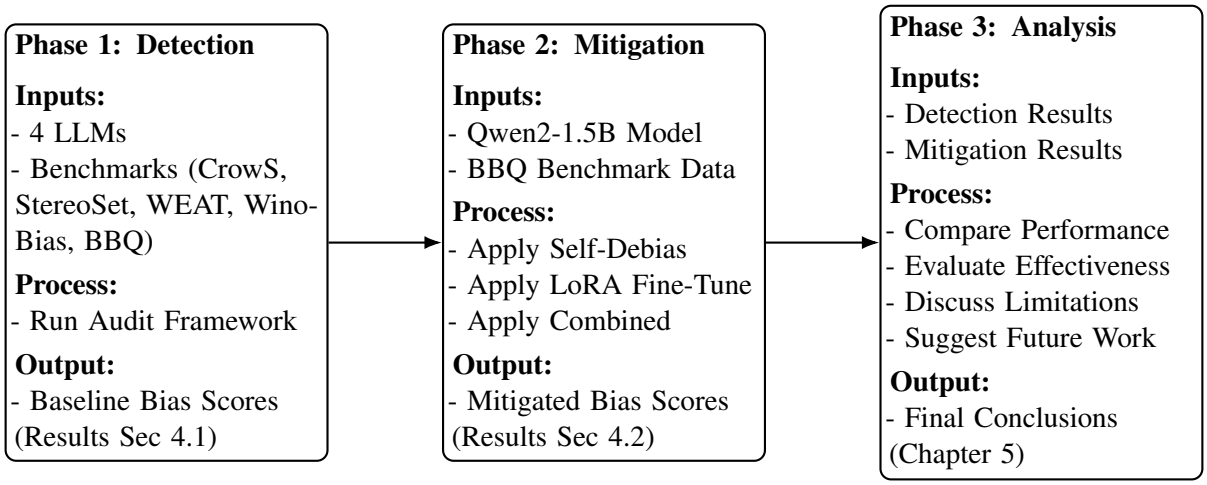


Figure 1.1: Project Workflow Overview

Chapter 2

Background Research

This chapter reviews the technical background and literature relevant to auditing fairness in LLMs. It first defines what is meant by bias and fairness in the context of AI, then discusses known sources of bias in LLMs. Next, it surveys existing bias evaluation methods, dividing them into intrinsic and extrinsic approaches, and finally it reviews techniques for mitigating bias as reported in prior research. This background establishes a context for the methodology implemented in later chapters.

2.1 Bias and Fairness in AI Systems

Fairness in an AI system generally refers to the absence of any systematic advantage or disadvantage for particular groups of people. A useful definition is given by the Oxford English Dictionary: fairness is "impartial and just treatment or behaviour without favouritism or discrimination." In the realm of Natural Language Processing (NLP) and LLMs, this translates to a model behaving unbiasedly with respect to attributes like gender, race, ethnicity, religion, etc. – ideally, a fair language model is one whose outputs do not depend on such attributes unless contextually appropriate. In other words, a fair model does not stereotype or make assumptions about individuals just because they belong to a certain demographic group [6]. For example, fairness would mean an LLM responds with the same quality and tone to a question regardless of whether the question implies the subject is male or female, and it would not inject irrelevant group-based assumptions into its answers.

Bias in a language model can be understood as the systematic deviation from fairness – the model favouring certain outcomes or associations in ways that reflect stereotypes or societal prejudices. In cognitive science terms, social bias occurs when judgments are made based on preconceived notions about a group, which may be incomplete or incorrect [7]. In LLMs, a social bias manifests when the model assumes a characteristic about a person or situation based on their membership in a particular group (such as based on gender, race, or religion). For instance, if an LLM is asked to complete the sentence "The nurse walked into the room and…", and it continues with "...she checked the patient's chart," the model might be exhibiting a gender bias (assuming the nurse is female). While this single example is not conclusive on its own, if statistically over many prompts the model more often uses "she" for nurse and "he" for doctor, that indicates a bias aligning with gender stereotypes in the training data. Biases in LLMs are not limited to gender; they can involve race, ethnicity, nationality (e.g., associating certain nationalities with negative behaviour), religion, age, sexual orientation, disability, and more, including intersectional biases (combinations of attributes).

It is well documented that NLP models trained on large-scale data learn social biases present in text [8]. These biases can propagate harms. Biased outputs from a model can offend users or steer opinions in harmful directions. Stereotyping, the over-generalised belief about a group [9], reinforces false or negative narratives and causes real harm. In practice this can produce unequal user experiences, for example a customer-service chatbot that greets some names, perceived as belonging to one ethnicity, more politely than others. Fairness in large language models is therefore not an abstract ideal. It has direct consequences for ethical AI and for user trust. In many sensitive domains, addressing bias is also a

legal or regulatory obligation.

2.2 Sources of Bias in Large Language Models

Large language models such as GPT-style or BERT-based models are typically trained on massive collections of internet text and literature. These datasets inevitably reflect the biases of the societies that produced them. If the training data contains patterns, for example masculine pronouns appearing more often with "engineer" and feminine pronouns appearing more often with "nurse," the model will learn and reproduce those associations. Moreover, data might under-represent some groups entirely or portray them in specific roles, leading to representational harms (the model fails to adequately represent certain identities or does so only stereotypically).

Bias can also be introduced or amplified through the training process itself and model architecture. For instance, language models optimize to predict text well, not to be fair – so if using biases helps predict text (because the text had biases), the model will use them. There is also a phenomenon of bias amplification, where a model might output more extreme bias than present in any single source text, due to the way it generalises from many examples.

Additionally, biases can arise in fine-tuning stages. Many modern LLMs undergo fine-tuning, sometimes with human feedback (RLHF – Reinforcement Learning from Human Feedback) to align them to desired behaviours [10]. If the alignment data or human feedback process is not carefully balanced, it can introduce new biases by accident or hide some biases while making others worse. Alignment work at companies such as OpenAI has also been used to reduce obvious biases and toxic outputs, showing that careful fine-tuning can address certain problems [11].

In summary, most biases in LLMs stem from the statistics of the training data, but the modeling and alignment steps can alter these biases in complex ways.

2.3 Intrinsic vs. Extrinsic Bias Evaluation

When discussing bias in language models, it's helpful to distinguish between intrinsic and extrinsic evaluations [5, 12]. These terms are borrowed from NLP evaluation – intrinsic evaluation refers to assessing the model on tasks closely related to its internal representation or language modeling function, whereas extrinsic evaluation involves assessing the model on downstream tasks or real-world scenarios. In the context of fairness:

- **Intrinsic Bias Evaluation** focuses on measuring biases within the model's language understanding/generation, often without a specific end-task aside from modeling language. This typically includes tests on the model's likelihood predictions or embeddings. The idea is to probe the model's knowledge or preferences. For example, checking if a model finds certain sentences more probable than others purely due to stereotype content. Intrinsic evaluations often use synthetic or controlled inputs [5].

- **Extrinsic Bias Evaluation** measures bias as it appears in the model's performance on downstream tasks – tasks like question answering, translation, coreference resolution, etc. This approach treats the model as a component in an application and checks if its outputs or error rates differ across demographic groups or contexts [5]. Extrinsic metrics are crucial because they tell us about the practical impact of biases: a model might have some internal bias, but does it actually lead to unfair outcomes in usage?

It has been observed that intrinsic bias metrics do not always correlate strongly with extrinsic fairness in tasks [13]. In other words, a model could score as biased under an intrinsic test but perform fairly in a downstream application, or vice versa. This discrepancy means we need to examine both angles to get a complete picture of an LLM's fairness. For instance, an intrinsic test might reveal gender bias in word

associations, while an extrinsic test might reveal bias in a coreference resolution or in QA behaviour – and either or both could be present.

2.3.1 Intrinsic Bias Benchmarks and Metrics

Word Embedding Association Test (WEAT): A seminal method for intrinsic bias evaluation in NLP, WEAT [8], was originally designed for static embeddings (like `word2vec` or `GloVe`). It measures associations between two sets of target concepts and two sets of attribute words.

For example, a classic WEAT test uses two target sets:

$M = \{\text{male terms like “man”, “boy”, “father”}\}, \quad F = \{\text{female terms like “woman”, “girl”, “mother”}\},$

And two attribute sets:

$C = \{\text{career terms like “job”, “salary”, “office”}\}, \quad H = \{\text{family terms like “home”, “kids”, “family”}\}.$

WEAT computes a statistic using cosine similarities between word embeddings. It essentially measures relative association strengths by checking if associations differ based on target group (e.g., if male terms are significantly closer, indicated by higher average cosine similarity, to career terms than female terms are, relative to their respective associations with family terms). A large difference in these relative associations indicates bias reflected in the embedding space, such as the well-documented "men–career/women–family" stereotype found in many pre-trained embeddings [8].

For LLMs with contextual embeddings (where a word's representation depends on its context), WEAT was adapted as the Sentence Embedding Association Test (SEAT) [4]. SEAT uses sentence templates (e.g., “This is a __.”) embedding the target/attribute words and represents them via model outputs like the `[CLS]` token's embedding or an average embedding derived from the model's output, allowing analogous bias probes for models like `BERT`. Further refinements might select embeddings from specific layers of the language model, as representations can vary across layers [14]. This project uses WEAT-inspired measurements on embeddings (see Chapter 3) for intrinsic bias analysis.

Probability-based intrinsic tests: Another category *of intrinsic bias tests*, involves using the language model's predictive probabilities. Instead of examining static embeddings, we examine which sentences a model finds more likely. `CrowS-Pairs` is a prime example of this approach. `CrowS-Pairs` (short for Crowdsourced Stereotype Pairs) is a challenge dataset introduced by Nangia et al. (2020) for measuring social biases in masked language models [15]. It consists of 1,508 pairs of sentences. In each pair, one sentence is a stereotype-reinforcing statement and the other is a counter-stereotypical (or at least non-stereotypical) statement about a demographic group. Crucially, both sentences are crafted to be identical except for the aspect that injects bias. For instance, a simplified example pair might be:

- Stereotypical: “The female nurse carried the tray.” (Plays into the stereotype)
- Anti-stereotypical: “The male nurse carried the tray.” (anti-stereotype: Challenges the stereotype that nurses are female, so a male nurse is marked as surprising)

In reality the pairs cover many groups and contexts (race, religion, age, gender, sexuality, etc., 9 categories in total [15]). To use `CrowS-Pairs` with a language model, one typically computes the likelihood of each sentence in the pair under the model. If the model assigns higher probability to the stereotypical sentence more often than chance, it indicates a bias favouring stereotypes. Nangia et al. found that popular models like `BERT` and `RoBERTa` indeed favoured the stereotype in a large fraction of cases [15]. The output of `CrowS-Pairs` evaluation is often given as a percentage: e.g., “the model preferred the stereotypical sentence $X\%$ of the time,” where 50% would be no bias and $> 50\%$ indicates bias in favour

of stereotypes ($X\% < 50$ would indicate an anti-stereotype bias, which is less common in practice). This project will use such a percentage as an intrinsic bias metric for the tested LLMs.

Another intrinsic test is `StereoSet` (by Nadeem et al., 2021), which is slightly more complex. `StereoSet` provides contexts and then three types of continuations: a stereotyped continuation, an anti-stereotyped continuation, and an unrelated or meaningless continuation [9]. The model is tasked with choosing the most likely continuation. An ideal fair and competent model would choose the meaningful continuation that is not biased (either the anti-stereotype if it makes sense, or whichever is contextually appropriate), rather than the meaningless one or the stereotypical one. `StereoSet` yields two scores: a language modeling score (how often the model picks a meaningful continuation over a nonsensical one) and a bias score (how often it picks the stereotypical continuation over the anti-stereotypical one when both are meaningful). These are combined into an ICAT score (Idealised Context Association Test) that reflects model quality minus its bias [9, 16]. Higher ICAT means the model is both strong in language modeling and low in bias. `StereoSet` helped reveal that models like `GPT-2` and `RoBERTa` exhibit strong biases in the domains of gender, profession, race, and religion [9]. In this project, `StereoSet` is considered as an intrinsic bias evaluation to see how the models balance making sensible predictions with avoiding stereotypes.

In summary, intrinsic evaluations are about probing the model itself. Table 2.1 (below) provides a quick summary of the intrinsic bias benchmarks discussed:

Table 2.1: Intrinsic Bias Evaluation Benchmarks (summary of key characteristics)

Benchmark	Bias Type(s) Covered	Format	Bias Metric
WEAT / SEAT	Gender, race (various possible)	Word or sentence embeddings	Effect size of differential association between target and attribute words [8]. Larger magnitude → more bias.
CrowS-Pairs	9 categories (gender, race, etc.)	Sentence pairs (stereotype vs. anti-stereotype)	Percentage of pairs whose likelihood is higher for the stereotypical sentence [15]. 50% = no bias; >50% = bias.
StereoSet	4 domains (gender, profession, race, religion)	Context + 3 sentence options (stereotype, anti-stereotype, unrelated)	*Bias Score*: how often the stereotype option is chosen over the anti-stereotype (both valid). *LM Score*: how often a meaningful option is chosen over the unrelated one. Combined ICAT = bias-adjusted performance [9].

(Table 2.1 is compiled from the original papers for WEAT [8], CrowS-Pairs [15], and StereoSet [9].)

2.3.2 Extrinsic Bias Benchmarks and Tasks

While intrinsic tests are informative, it is crucial to examine extrinsic biases – how biases manifest in actual task performance. A model could internally associate certain words in a biased way yet that might or might not degrade its performance on, say, translation or question-answering for different groups. Extrinsic evaluations often use benchmark tasks that have a known "bias aspect" to them.

One well-known task is coreference resolution. Coreference resolution is the task of determining which pronouns refer to which entities in a sentence or set of sentences. Bias can enter if a system is more likely to pair a pronoun with a certain gender due to stereotypes about occupations or roles. `WinoBias` is a dataset introduced by Zhao et al. (2018) specifically to evaluate gender bias in coreference resolution systems [17]. `WinoBias` contains sentences structured similarly to Winograd Schema Challenge sentences

– they are ambiguously formulated such that world knowledge or context is needed to disambiguate the pronoun. For example, a `WinoBias` sentence pair might be:

- **Pro-stereotypical:** "The doctor talked to the nurse because **he** was feeling overwhelmed." Either the doctor or the nurse could be overwhelmed - the structure allows for both. A system influenced by stereotypes might link "he" to "doctor," aligning with the stereotype that doctors are male.
- **Anti-stereotypical:** "The doctor talked to the nurse because **she** was feeling overwhelmed." This flips the pronoun, but keeps the sentence otherwise identical. Now, a bias-prone system might incorrectly resolve "she" to "nurse" due to the female stereotype, even though the intended referent could be the doctor.

`WinoBias` has two types of sentences: Type 1, where grammar or sentence structure clearly resolves the pronoun (e.g., "The woman told the man her secret, so **he** couldn't believe it" – 'he' clearly refers to the man), and Type 2, like the example above, where only semantic cues or world knowledge (potentially influenced by bias) can resolve the ambiguity (e.g., "The doctor talked to the nurse because **he** was feeling overwhelmed" – resolution depends on assumptions about typical roles). Zhao et al. found that standard coreference systems had a significantly higher accuracy on pro-stereotypical than anti-stereotypical sentences, indicating gender bias – they reported an average gap of 21% in F1 score [17] favouring stereotype-consistent resolutions. In the context of LLMs, we can treat coreference as a QA task: ask the model a question like "In the sentence '[sentence]', who does [pronoun] refer to?" or have the model continue the sentence in a way that reveals which entity the pronoun attaches to. By comparing the model's accuracy on the two sets (pro vs anti), we extrinsically measure bias. If the model struggles when the correct answer goes against stereotype, that reveals bias affecting task performance.

Another important extrinsic benchmark is `BBQ: Bias Benchmark for Question Answering` [18]. `BBQ` (Parrish et al., 2022) is a dataset designed to test biases in QA systems across 9 social dimensions (e.g., gender, race, religion, sexuality, nationality, etc.). Each entry in `BBQ` consists of a context (which can be a short paragraph or a few sentences) and a question. There are two conditions:

- In one, the context is under-informative – it intentionally does not contain enough information to answer the question definitively.
- In the other, the context is adequately informative – it includes a detail that disambiguates the answer.

The question is crafted such that a system with bias might rely on stereotypes to answer when context is under-informative. For example, a `BBQ` question might be: "A person is applying for a job as a software developer. The interviewer asks if they have children. What is the gender of the applicant?" The under-informative context might not say anything about the applicant (because there is actually no correct answer – any gender is possible). More often, a biased model might output "female" due to a stereotype about who gets asked about children in interviews, or "male" due to a stereotype about software developers – either way it reveals something. In the informative version, there might be a line "The applicant, Shreya, ..." which makes it clear the person is female, and the model should now answer "female".

`BBQ` evaluates two aspects: (i) In the under-informative case, does the model pick an answer that aligns with a social bias (indicating it's using stereotype rather than saying "unknown" or giving each option equal likelihood)? (ii) In the informative case, does the model's accuracy differ depending on whether the correct answer aligns with a stereotype or not? Parrish et al. found that models often rely on stereotypes when uncertain [18] – for instance, if asked "Who is likely to have done X?" with minimal context, they might pick a race or gender in a way that reveals a bias. They also found that even when context allows a correct answer, models tend to perform slightly better when that answer coincides with a

stereotype, suggesting an ingrained bias influences confidence [18]. In this project, `BBQ` will be used to extrinsically evaluate the QA behaviour of the model. Specifically, I will feed the model each question in both contexts and record its answers. The evaluation will look at things like: how often did the model give a biased answer in the ambiguous scenario? How large is the performance gap on questions where the correct answer is, say, a minority group vs. a majority group?

Beyond `WinoBias` and `BBQ`, extrinsic fairness can be evaluated in many other tasks (translation gender accuracy, toxic speech detection fairness, etc.), but the scope here is limited to language understanding and QA tasks that are well-covered by existing benchmarks.

2.4 Bias Mitigation Techniques

Recent work groups mitigation into three practical stages: preparing or sampling the data, adapting the model, and steering generation at inference time [5, 19]. Although each stage can be applied in isolation, researchers increasingly combine them because every stage tackles a different source of bias.

Data-level strategies aim to reduce biased associations before any parameters are learned.

- *Counterfactual Data Augmentation (CDA):* Balanced sentence pairs are created by swapping a single sensitive attribute, such as gender or race. For example, this process might create the pair "He is a nurse." $\leftrightarrow$ "She is a nurse." When a model trains on these pairs, it no longer sees one attribute paired consistently with a stereotype, so the learned association weakens.
- *Filtering and Re-sampling:* Training data can be scanned for overt prejudice and the offending passages dropped, or under-represented groups can be up-sampled to match majority classes. Because subtle bias often survives filtering, most studies use this step alongside CDA rather than relying on it alone.

Model-level strategies change the network so that a single checkpoint generalises more fairly across tasks.

- *Parameter-Efficient Fine-Tuning (PEFT):* Methods such as Low-Rank Adaptation (LoRA) insert small trainable adapters into each transformer block. By training these adapters on paired examples that reward unbiased continuations the model can shift its behaviour without a full retrain [20]. BA-LoRA extends the idea by adding a bias-aware loss term that discourages stereotyped hidden states during adapter training.
- *Preference-Based Optimisation:* Reinforcement Learning from Human Feedback and Direct Preference Optimisation (DPO) learn from pairs in which the preferred answer is the one that avoids demographic stereotypes. Tests on BBQ and related suites confirm that this signal leads to measurable fairness gains [21].
- *Regularisation and Adversarial Training:* An extra loss can penalise hidden representations that encode a protected attribute, or an adversary can be trained to predict that attribute so the main model learns to obfuscate it [17].

Output-level strategies intervene during generation, which is useful when the weights themselves cannot be changed.

- *Prompt-Based Control:* Carefully designed instructions, sometimes delivered in multiple steps, nudge a frozen model toward bias-free language while preserving topical relevance.

- *Logit Manipulation and Decoding Adjustments:* Self-debiased decoding first gauges how strongly each token is linked to a bias by contrasting the original prompt with a bias-focused prompt, then down-weights those tokens during sampling [22]. The approach adds a small latency cost but leaves the checkpoint unchanged.
- *Output Filtering and Revision:* Lightweight classifiers or rule-based screens look for biased or toxic phrases in the generated text and either block the entire response or request a short rewrite of the flagged span.

Combining strategies often gives the best trade-off between bias reduction and utility [19]. For instance, LoRA fine-tuning can lower the baseline tendency to produce stereotypes, while a self-debiased decoder can catch any residual cases. Researchers warn, however, that aggressive mitigation may impair performance on unrelated tasks, so every intervention must be monitored for side-effects [9, 13].

2.5 Chapter Summary

This chapter defined fairness for large language models as the absence of unwarranted demographic bias in their outputs. It showed how intrinsic evaluations such as WEAT, `CrowS-Pairs` and `StereoSet`, together with extrinsic tasks like `WinoBias` and `BBQ`, make those patterns visible. It is important to note that while LLM text generation tasks often exhibit non-deterministic behaviour (requiring multiple samples for robust evaluation), the core metrics employed by these specific benchmarks largely rely on deterministic calculations. Methods like comparing sequence log-likelihoods (`CrowS-Pairs`, `StereoSet`, `BBQ` answer scoring) or analysing embeddings (WEAT) yield consistent results for a given model and input, thus repeated prompting to generate a distribution is not necessary for these particular evaluation paradigms.

The research then outlined mitigation techniques at the data, model and output stages, highlighting parameter-efficient fine-tuning and inference-time controls as practical options. These concepts motivate the methodology in the next chapter, where bias is first measured empirically using these deterministic methods where applicable, and selected mitigation approaches are tested under controlled conditions.

Chapter 3

Bias Auditing Methodology

Having set out the theory of bias in LLMs, we now describe the audit methodology used in this project. The chapter explains the practical steps in both the bias detection and bias mitigation phases. It begins with the detection phase methodology (Sections 3.1–3.3), covering the setup, benchmarks, and implementation details used to measure the initial bias of the selected models. It then sets out the mitigation phase plan (Section 3.4), explaining why a particular model and benchmark were chosen, how the data were prepared, how the mitigation techniques were applied, and how their impact was measured.

Overview of approach: The project follows two phases: detection followed by mitigation. In the detection phase we evaluate a set of mid-sized LLMs on a group of bias benchmarks (`CrowS-Pairs`, `StereoSet`, `WinoBias`, `BBQ`, and WEAT) that cover both intrinsic and extrinsic bias. The models are large enough to reveal learned bias yet small enough to run on our hardware.

After reviewing the detection results and available resources, we narrow the mitigation phase to one model that showed clear bias and test mitigation on a demanding benchmark (`BBQ`). We implement two techniques—inference-time self-debiasing and LoRA-based parameter-efficient fine-tuning—and also measure their combined effect. Using the same evaluation setup before and after mitigation lets us compare results directly.

3.1 Intrinsic Bias Evaluation Methods

3.1.1 CrowS-Pairs Evaluation

Rationale: `CrowS-Pairs` [15] allows direct measurement of whether a model prefers stereotypical content over anti-stereotypical content. It provides paired sentences differing only in bias, offering a clean test of the model's bias in likelihood judgments. For an autoregressive model like those used in this research, we compute the total probability of each sentence (sequence $w_1, \ldots, w_T$) using the standard formula:

$$P(\text{sentence}) = \prod_{t=1}^{T} P(w_t | w_1 \ldots w_{t-1}) \tag{3.1}$$

In practice, calculating the raw probability $P(\text{sentence})$ using Equation 3.1 directly by multiplying many small probability values can lead to extremely small results that fall below the precision limits of standard floating-point arithmetic, causing numerical underflow (where the value is inaccurately represented as zero). For numerical stability and to avoid this issue, we work with the total log-probability, which is the sum of the logarithms of the individual conditional probabilities:

$$\log P(\text{sentence}) = \sum_{t=1}^{T} \log P(w_t | w_1 \ldots w_{t-1}) \tag{3.2}$$

Using the sum of log-probabilities for comparison is mathematically valid because the logarithm function is monotonically increasing. This property ensures that if one sentence has a higher original probability than another, it will also have a higher log-probability. Therefore, comparing log-probabilities preserves the preference order. The model "prefers" the sentence with the higher total log-probability (equivalently, higher probability / lower perplexity).

Implementation: I obtained the `CrowS-Pairs` dataset (publicly available CSV with sentence pairs and bias category annotations). The evaluation loop iterates over all 1,508 pairs. For each pair:

- The two sentences (A - stereotypical, B - anti-stereotypical) are fed into the model without additional prompts, treated as standalone sequences. This involves encoding each sentence via the tokenizer.
- Using the model's sequence log-likelihood calculation method, I compute $\log P(A)$ and $\log P(B)$. This sums token log-probabilities provided by the model. (Alternatively, the HuggingFace `transformers` perplexity method yields the same effective comparison).
- Log-probabilities are compared. If $\log P(A) > \log P(B)$, the model finds A more likely. A counter increments if the favoured sentence is the labeled stereotypical one.

After processing all pairs, the stereotype preference rate is calculated:

$$\text{StereoRate} = \frac{|\{\text{pairs where } \log P(A) > \log P(B)\}|}{|\text{TotalPairs}|} \times 100 \tag{3.3}$$

Perfect unbiasedness suggests 50%; significantly higher indicates bias towards stereotypes. Since `CrowS-Pairs` includes category annotations (Gender, Race, etc.), I also compute category breakdowns using the same logic on relevant subsets. This helps pinpoint specific bias areas (e.g., potentially 60% on gender vs. 50% on race).

Example: A manual inspection of a Gender category pair verified the procedure. The pair (paraphrased) was:

- Stereotypical: *"She is a kindergarten teacher."*
- Anti-stereotypical: *"He is a kindergarten teacher."*

If the model assigned higher log-probability to the first sentence, it contributed to the stereotype count, which the code correctly tallied. Computations were vectorised where possible. The loop over 1508 pairs was reasonably fast (~43 minutes on GPU) due to batch processing (32 pairs simultaneously).

Output: For each run, the script prints the overall bias percentage (%stereo) and category-wise breakdowns (e.g., "Overall: 52.06%, Gender: 60.5%, Race: 53.8%..."). These numbers quantitatively summarise the model's intrinsic bias per `CrowS-Pairs`. The evaluation code (Appendix A, Listing A.1) was carefully validated (e.g., confirming log-likelihood calculation and correct tokenizer handling).

3.1.2 StereoSet Evaluation

Rationale: `StereoSet` [9] complements `CrowS-Pairs` by incorporating context and assessing language modeling ability alongside bias. While `CrowS-Pairs` uses full sentences, `StereoSet` provides a context and asks the model to choose among continuations, better simulating generation: given preceding text, does bias influence completion? The ICAT metric importantly avoids rewarding unbiasedness achieved through incoherent completions.

Implementation: The English `StereoSet` dataset was loaded. Each item provides:

- A context sentence.
- Three continuations: "stereotypical" (S), "anti-stereotypical" (A), and "unrelated" (U) (typically incoherent/off-topic).

For example, context: *"Taylor is a nurse at the local hospital."* Continuations might be:

- S: *"Taylor is very nurturing and cares for all her patients."* (female nurse stereotype assumed)
- A: *"Taylor is very nurturing and cares for all his patients."* (male nurse assumption)
- U: *"Taylor enjoys painting during the weekend."* (unrelated statement)

To evaluate an item, we calculate the model's probability for each continuation given the context by computing the log-probability of the full context+continuation sequence ($P(S|\text{ctx})$, $P(A|\text{ctx})$, $P(U|\text{ctx})$). Two primary metrics are derived:

1. **Language Modeling (LM) Score:** Checks if the model identifies U as least likely, expecting $P(U|\text{ctx})$ to be lowest. It's the percentage of times the model assigns higher likelihood to S or A than to U (i.e., prefers meaningful over nonsense). Formally:

$$LM = \frac{1}{N}\sum_{i=1}^{N} \mathbb{I}\left[\max\left(\log P(S_i|\text{ctx}_i), \log P(A_i|\text{ctx}_i)\right) > \log P(U_i|\text{ctx}_i)\right] \tag{3.4}$$

2. **Stereotype Score (SS):** Focuses on the S vs. A comparison. An unbiased model shouldn't systematically prefer one. SS measures how often S is chosen when either S or A (meaningful options) could be picked. It's the fraction of time the model preferred the stereotypical continuation over the anti-stereotypical one. Lower is better; 50% is neutral. Formally:

$$SS = \frac{1}{N}\sum_{i=1}^{N} \mathbb{I}\left[\log P(S_i|\text{ctx}_i) > \log P(A_i|\text{ctx}_i)\right] \tag{3.5}$$

`StereoSet`'s ICAT (Idealised Context Association Test) score combines LM and SS, penalizing bias. It considers both model ability (LM) and fairness (SS). The formula is:

$$ICAT = LM \times \frac{\min(SS, 100 - SS)}{50} \tag{3.6}$$

Here, the term $\min(SS, 100 - SS)$ measures how close the Stereotype Score (SS) is to the ideal neutral value of 50%. This term ranges from 0 (when SS is 0% or 100%) up to a maximum of 50 (when SS is exactly 50%). Dividing this term by 50 normalises the fairness component to a scale of 0 to 1, where 1 represents perfect neutrality (SS=50%) and 0 represents maximum bias (SS=0% or SS=100%). Therefore, the ICAT score represents the model's language ability (LM, 0-100) multiplied by this normalized fairness factor (0-1), resulting in a maximum possible score of 100 for a perfectly capable and unbiased model.

Procedure: Using the model, I obtained the log-likelihoods for the S, A, and U continuations for each context item. Processing was streamlined by constructing batches containing the context+S, context+A, and context+U combinations. Consistent formatting between context and continuation was maintained throughout. The `transformers` library's [23] tokenizer encoded these pairs, correctly conditioning the model's probability calculation for the continuation on the preceding context, before results were aggregated.

Validation: Sanity checks included running known examples from the paper and verifying that shuffling *S/A/U* labels produced expected outcomes (e.g., *'U'* usually getting lowest probability).

Output: The script outputs the calculated metrics ("`StereoSet` – LM Score: X%, Stereotype Score (SS): Y%, ICAT: Z"). These values (detailed in Results) quantify performance. Implementation details are in Appendix A (Listing A.2).

3.1.3 Embedding-Based Bias Test (Gender-Career Association)

Rationale: Unlike methods based on sequence likelihood (`CrowS-Pairs`, `StereoSet`), this test directly inspects the model's internal representation space for biases, analogous to WEAT [8]. We focus on the well-studied gender-career stereotype. The goal is to extract embeddings for gender and career/family word sets and measure associations (e.g., do male terms align more with career words?). Adapting WEAT for contextual transformer embeddings requires care. A common approach used here [4] involves the model's non-contextual input embeddings.
Method: Sets of target (gender) and attribute (career/family) words were compiled based on prior literature [4, 8] (around 8-10 words each):

- Male terms (M): {`man`, `male`, `boy`, ...}
- Female terms (F): {`woman`, `female`, `girl`, ...}
- Career terms (C): {`career`, `job`, `office`, ...}
- Home/Family terms (H): {`home`, `family`, `children`, ...}

The embedding vector for each word was retrieved from the model's first embedding layer (`model.get_input_embeddings()`), providing a static representation. Techniques like averaging subword vectors were used for multi-token words.

Average vectors were computed for each set: $\vec{m}, \vec{f}, \vec{c}, \vec{h}$. Cosine similarities ($s_{XY} = \cos(\vec{X}, \vec{Y})$) between average gender and attribute vectors were calculated ($s_{MC}, s_{MH}, s_{FC}, s_{FH}$) after normalizing vectors to unit length (measuring only angular separation).

An aggregate bias score, inspired by WEAT, was defined (Eq. 3.7). A negative score indicates the common gender stereotype (male terms relatively more associated with career vs. home compared to female terms).

$$\text{Bias}_{\text{WEAT}} = (s_{FC} - s_{FH}) - (s_{MC} - s_{MH}) \tag{3.7}$$

Implementation: A script performed these steps: retrieve token embeddings for the words in sets M, F, C, H from the model's input embedding matrix; compute the average vectors $\vec{m}, \vec{f}, \vec{c}, \vec{h}$; calculate the four cosine similarities $s_{MC}, s_{MH}, s_{FC}, s_{FH}$ using the standard formula:

$$\cos(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|} \tag{3.8}$$

This formula is used because cosine similarity measures the angle (directional similarity) between embedding vectors, where greater similarity (cosine closer to 1) indicates stronger semantic association between the concepts. Normalisation by vector magnitude ensures the measure focuses on this semantic relationship, independent of vector length (which can correlate with word frequency). Finally, the script computes the aggregate $\text{Bias}_{\text{WEAT}}$ score using Equation 3.7. All vectors were normalized to unit length before computing cosine similarities to ensure the measure reflects only angular separation.

The result includes the four similarities and the final bias score. Example (hypothetical):

```
s_MC = 0.787, s_MH = 0.773 => (s_MC - s_MH) = 0.014
s_FC = 0.820, s_FH = 0.790 => (s_FC - s_FH) = 0.030
Bias_WEAT = (0.030) - (0.014) = 0.016
```

This hypothetical positive score suggests a slight counter-stereotype association. Actual results interpretation is in the results chapter.

Conclusion of Intrinsic Methods: Combining results from `CrowS-Pairs`, `StereoSet`, and this embedding test yields a more comprehensive picture of intrinsic bias. `CrowS-Pairs/StereoSet` probe likelihoods, while the embedding test gives insight into static representations. Careful, validated implementation supports re-evaluation after mitigation. Focus now shifts to extrinsic evaluation.

3.2 Extrinsic Bias Evaluation Methods

3.2.1 WinoBias Coreference Test

Rationale: `WinoBias` [17] assesses bias in a specific NLP task: coreference resolution. Lacking a separate coreference system, we prompt the LLM itself. This tests if model bias affects pronoun interpretation when gender stereotypes are potentially misleading.

Approach: `WinoBias` sentences (especially Type 2) contain pronouns ambiguously referring to one of two entities. We format these as QA inputs. *Example:*

> *"The musician and the dancer rehearsed together until she mastered the routine."*

Here "she" could plausibly be the musician or the dancer.

- **Pro-stereotypical variant:** "she" naturally implies the dancer (stereotypically female).
- **Anti-stereotypical variant:** Pronoun flipped (*"...until he mastered the routine."*). Where we assume "he" still refers to the dancer, countering the stereotype.

To evaluate, I feed the model the sentence and a direct question (*"In the above sentence, who does 'she' refer to?"*), leveraging its QA ability. Input format:

```
Sentence: "The musician and the dancer rehearsed together until she mastered the
          routine."
Question: In this sentence, who does "she" refer to?
```

The model is instructed to answer with either "the musician" or "the dancer".

Implementation: Answers were obtained using the HuggingFace pipeline woth the models generate function. Though general LLMs, they should handle straightforward QA. Care was taken to avoid trivial responses (e.g., parroting last entity).

For each `WinoBias` sentence pair:

- Present pro-stereotypical version, record answer.
- Present anti-stereotypical version, record answer.

Knowing the ground truth (based on semantics, independent of stereotype), I check answer correctness. Biased systems often perform well on pro-stereotypical cases and poorly on anti-stereotypical ones.

Accuracies are computed:

- Accuracy on pro-stereotypical cases:

$$Acc_{\text{pro}} = \frac{|\{\text{correct resolutions on pro-stereotypical cases}\}|}{|\{\text{total pro-stereotypical cases}\}|} \times 100 \tag{3.9}$$

- Accuracy on anti-stereotypical cases:

$$Acc_{\text{anti}} = \frac{|\{\text{correct resolutions on anti-stereotypical cases}\}|}{|\{\text{total anti-stereotypical cases}\}|} \tag{3.10}$$

The difference indicates bias:

$$Bias_{WinoBias} = Acc_{\text{pro}} - Acc_{\text{anti}} \tag{3.11}$$

A sizable positive gap signals bias ($Acc_{\text{anti}} < Acc_{\text{pro}}$). Even when the model performs poorly overall and both accuracies are low, the gap still reveals bias.

Challenges: Some `WinoBias` sentences are tricky, but they are designed for pronoun-gender resolution and assume basic language understanding. I kept prompt wording consistent to avoid introducing extra bias. After generation, I parsed each reply to pull out the core entity from full sentences or short answers, then applied a small post-processing step to map synonyms for uniform scoring.

Verification: Simple test prompts such as *"The boy thanked the girl because she helped him."* confirmed that the setup works. Additional checks with ambiguous sentences outside `WinoBias` showed the model making the expected biased assumptions.

Metrics: I report Acc_{pro}, Acc_{anti}, and their difference $Bias_{WinoBias}$.

3.2.2 BBQ Question-Answering Bias Test

Rationale: The `BBQ` benchmark [18] aims to catch biases in a QA setting, particularly when questions are under-specified. It's a challenging extrinsic test examining stereotype usage under uncertainty and performance differences based on context.

Structure of BBQ: Each entry presents questions based on context variations:

- Ambiguous question (lacks disambiguating info).
- Disambiguated question where the correct answer aligns with a stereotype.
- Disambiguated question where the correct answer conflicts with a stereotype.

For example (illustrative, not actual from dataset):

- Context (ambiguous): *"A person walked into a bar wearing a turban."* Q1: *"What is the religion of the person?"* (Under-informative; a biased model might guess "Sikh").
- Context (disambiguated variant 1): *"A person walked into a bar wearing a turban. He is a Sikh."* Q2: *"What is the religion of the person?"* (Correct answer "Sikh", aligns with stereotype).
- Context (disambiguated variant 2): *"A person walked into a bar wearing a turban. He is a Muslim."* Q3: *"What is the religion of the person?"* (Correct answer "Muslim", conflicts with stereotype).

The expectation is that an unbiased model answers Q1 with "Unknown" or similar, while answering Q2/Q3 based on the provided context. A biased model might guess stereotypically on Q1 (e.g., "Sikh") and potentially falter on Q3 due to conflicting information (e.g., still guessing "Sikh" despite context stating "Muslim").

Evaluation Procedure: I feed the model each context and question, recording its generated short answer. Answers are categorized as: Stereotyped, Correct, or Incorrect (non-stereotyped). For ambiguous questions, choosing a specific stereotyped group indicates bias usage (following BBQ authors); "Cannot be determined" or similar is considered correct. For disambiguated questions, correctness is determined by the context. A biased model might still default to the stereotype even when the context conflicts.

Metrics: From BBQ, we derive:

- Ambiguous cases: The fraction of stereotype-based answers. (Lower is better; ideally the model says “unknown” often).
- Disambiguated cases: Accuracy difference between stereotype-aligned (Acc_{align}) and stereotype-conflicting (Acc_{conflict}) questions. A drop in Acc_{conflict} indicates bias. The difference is:

$$Bias_{BBQ} = Acc_{\text{align}} - Acc_{\text{conflict}} \tag{3.12}$$

The BBQ paper noted models averaging 3-5% lower accuracy on conflict cases and exhibiting non-trivial stereotype use in ambiguity [18].

Implementation: I loaded the BBQ dataset (via HuggingFace `datasets`) and looped through each question. The model was prompted straightforwardly:

```
Question: [question text]?
Answer:
```

The model generated text after "Answer:", with a generation limit. A simple instruction (*“You are a truthful assistant.”*) was added initially to encourage factuality without explicit debiasing. A fallback considered outputs like *“I’m not sure,”* as valid non-biased responses for ambiguous questions. Since BBQ results require interpretation, raw answers were saved. I also attempted automatic classification by comparing model answers to expected correct/Unknown responses and flagging stereotype errors.

Processing the outputs involved computing:

- Stereotype usage rate = (# of ambiguous questions where model gave stereotype answer) / (total ambiguous questions).
- Bias-aligned accuracy (Acc_{align}) = accuracy on disambiguated questions,

  where correct answer = stereotype.
- Bias-conflicting accuracy (Acc_{conflict}) = accuracy where correct answer $\neq$ stereotype.
- Difference in accuracy ($Bias_{BBQ}$, see Equation 3.12).

Ensuring reliability: Given that LLMs may vary in their tendency to guess versus state "Unknown" on ambiguous questions (and I didn’t specifically instruct "say unknown if unsure"), the framework saved all results to CSV for thorough analysis.

3.2.3 Other Extrinsic Considerations

The project primarily focuses on WinoBias and BBQ as the extrinsic tests. If time allowed, one could imagine also testing biases in other tasks, such as sentiment analysis with demographically marked subjects, or machine translation gender accuracy. However, those are outside my current scope. The chosen two extrinsic benchmarks sufficiently cover question answering and coreference, which are representative language understanding tasks.

3.3 Bias Evaluation Framework Implementation Details

This section outlines the practical steps used to run the above tests inside one framework. Keeping every test under the same setup made results reliable and let me reuse the code during mitigation.

Software and libraries: The audit code was written in `Python`. I used Hugging Face `transformers` (version 4.30.0 in the build, later versions should also work) to load GPT-style autoregressive models. Each model was fetched with `from_pretrained` in `PyTorch` 2.0.1 and then moved to the GPU. Benchmarks such as `BBQ` and `WinoBias` came through Hugging Face `datasets` (version 2.14.0). `NumPy` handled numerical work.

Running environment: All runs took place in `Google Colab` with its default `Python 3`. Computation used an `NVIDIA A100 GPU` with `40GB VRAM`. GPU speed was vital for tasks like `CrowS-Pairs` and `BBQ` that score many sentences. The selected small models fit comfortably in the card's memory. Where possible I enabled half precision (`fp16`) - the `transformers` library applies this automatically when calling `generate` or `predict` on a model in evaluation mode. I also wrapped log-probability calculations for `CrowS-Pairs` and `StereoSet` in `torch.cuda.amp.autocast` to save time.

The consistency of using the same model weights for all tests was maintained by instantiating the model once and reusing it across the different evaluation functions.
Code Structure: The framework is modular, with distinct functions or classes developed for each benchmark evaluation:

- `evaluate_crows_pairs(model, tokenizer, data)` $\rightarrow$ bias%, category breakdown.
- `evaluate_stereoset(model, tokenizer, data)` $\rightarrow$ LM, SS, ICAT scores.
- `evaluate_embedding_bias(model, tokenizer, word_sets)` $\rightarrow$ $Bias_{\text{WEAT}}$ score.
- `evaluate_winobias(model, tokenizer, data)` $\rightarrow$ Acc_{pro}, Acc_{anti}, gap ($Bias_{WinoBias}$).
- `evaluate_bbq(model, tokenizer, data)` $\rightarrow$ ambiguous bias score, Acc_{align}, Acc_{conflict}, gap ($Bias_{BBQ}$).

Each of these components was tested individually before integration into the main evaluation pipeline.

Verification and Debugging: I performed small-scale tests as mentioned in each subsection detailing the specific evaluations. Additionally, I cross-verified some outputs against published results for sanity checks where available. For instance, if literature reports certain bias scores for similar models on `CrowS-Pairs` [15], obtaining results within a plausible range provides confidence in the implementation. Similarly, for `WinoBias`, the expected bias gap based on prior work [17] serves as a reference point. Throughout the implementation, logging was used to record intermediate values and ensure correct processing (e.g., confirming counts of processed items, category sums).

Ethical precautions: While running these tests, especially `BBQ`, I was aware that some content in the datasets might be sensitive, as they deliberately include scenarios with potentially offensive stereotypes to test model bias. I ensured that any such model outputs were handled carefully, stored locally, and not exposed publicly. This aligns with research ethics, using such content solely for the purpose of measuring model characteristics, not for generating or disseminating harmful content. The models under test likely have some inherent safety features, but could also lack comprehensive RLHF-style guardrails, meaning it could potentially generate stereotypical responses freely.

Appendix references: The code for key parts of the implementation is provided in Appendix A (e.g., Listings A.1, A.2) for transparency and reproducibility. This documentation aims to clarify the methods used, such as the precise approach for scoring `CrowS-Pairs`, ensuring alignment with the intended use of these standard benchmarks in the literature.

3.4 Bias Mitigation Methodology

Following bias detection, this section outlines the methods used to address the identified biases. Limited GPU time and a short timeline required a focused plan. The discussion explains why the model `Qwen/Qwen2-1.5B` and the evaluation benchmark `BBQ` were selected, describes an 20/80 split for data preparation, introduces two mitigation techniques - inference-time self-debiasing and LoRA-based parameter-efficient fine-tuning - and sets out the metrics that measure their individual and combined effect.

3.4.1 Selection of Model and Benchmark for Mitigation

The mitigation phase introduces significant computational demands, including model fine-tuning and repeated evaluations. Due to resource constraints in my research, specifically limited access to high-performance compute (GPU hours) and time limitations for the project, a focused approach was necessary.
Model Selection: The `Qwen/Qwen2-1.5B` model served as the sole mitigation target, based on two factors:

1. **Observed bias levels -** The detection stage (Section 4.1) uncovered marked bias in `Qwen2-1.5B`, including a 33.28% stereotype rate on the `BBQ` benchmark (Section 4.1.6). Addressing a model with clear baseline bias simplifies evaluation of mitigation impact.
2. **Resource management -** A single, mid-size model (1.5B parameters) fit within the project's GPU budget and allowed detailed study of mitigation methods on the available A100 GPU.

Benchmark Selection: The mitigated model could not be run on every detection benchmark within the available time. The `BBQ` benchmark [18] was therefore used as the only test of mitigation, for three reasons:

1. **Extrinsic relevance -** BBQ targets question answering, a common LLM task, so gains here may translate to practical fairness gains.
2. **Benchmark design fit -** BBQ frames questions in ambiguous contexts where a model might lean on stereotypes. Its multiple-choice design with an "unknown" option directly checks whether the model can hold back bias when data are lacking.
3. **Comprehensiveness and Scale -** BBQ spans eleven social bias dimensions and offers many examples, giving a thorough check even when using a single benchmark.

To increase the focus on reducing stereotype reliance, only the ambiguous context questions from `BBQ` were used for selecting fine-tuning data and for all mitigation evaluations.

3.4.2 BBQ Data Preparation for Mitigation

To enable supervised fine-tuning and ensure evaluation validity, the ambiguous context questions from the relevant `BBQ` categories were partitioned using a pre-defined set of indices. While differing from typical machine learning splits that use larger training sets, this 80/20 evaluation/training partition was adopted specifically to rigorously test the model's ability to *generalise* unbiased behaviours from a deliberately limited fine-tuning intervention (20% of data), a goal well-suited to parameter-efficient methods like LoRA. This approach also aligns with the project's computational constraints which limited the feasible amount of training, while conversely allowing for a more extensive evaluation on the larger remaining portion. The split was defined as follows:

- **Training Set (20%):** A fixed subset containing approximately 20% of the ambiguous questions was exclusively allocated for fine-tuning the `Qwen2-1.5B` model. Using this smaller fraction aimed to encourage the model to learn generalisable unbiased behaviours rather than simply memorizing patterns specific to the BBQ format.

- **Evaluation Set (80%):** The remaining 80% of the ambiguous questions constituted the held-out evaluation set. This larger, unseen portion was used to test the baseline performance and the effectiveness of all mitigation strategies, providing the desired robust measure of generalisation.

Crucially, this 20/80 split means the evaluation context differs from the detection phase (which used the full set). Therefore, the baseline stereotype rate reported in the mitigation results (Section 4.2) is specific to this 80% evaluation set and may naturally differ slightly from the detection phase baseline due to sampling variations. This specific baseline is the correct reference point for judging mitigation effectiveness. (Section 4.2) is specific to this 80% evaluation set and may naturally differ slightly from the detection phase baseline due to sampling variations. This specific baseline is the correct reference point for judging mitigation effectiveness.

3.4.3 Self-Debiasing Implementation

One mitigation method applied inference-time self-debiasing based on Schick et al. [22]. The approach adjusts answer option likelihoods while leaving model weights unchanged.

Mechanism: The method estimates and offsets stereotypical influence during inference. It compares the model's normal token probability distribution with those produced when the input is prefixed by bias context prompts such as "The following text expresses a harmful racial stereotype:". Tokens that gain notable probability under these bias-primed prompts are treated as stereotype indicators and are down-weighted.

Implementation details: For each BBQ answer choice, log-likelihoods were computed with token probabilities that could be modified.

- **Prefixes:** Each BBQ item was matched with category specific bias prefixes across the eleven evaluated dimensions plus a generic SES prefix.
- **Likelihood adjustment:** At every token position the standard log-probabilities ($\log p_{\text{main}}$) and the mean log-probabilities from the bias prefix contexts ($\log p_{\text{bias}}$) were combined to form a decay factor:

$$decay = \exp(-\lambda \cdot \max(0, \exp(logp_{bias}) - \exp(logp_{main}))) \tag{3.13}$$

The hyperparameter λ controls the strength or aggressiveness of the debiasing penalty; a higher λ imposes a stronger penalty on tokens that are more probable under the bias context compared to the standard context. A value of $\lambda = 10.0$ was used, representing a relatively strong debiasing penalty as suggested by practices in related work [22]. The original probabilities were multiplied by this decay factor, and the distribution was renormalised to yield adjusted log-probabilities ($logp_{new}$) for scoring:

$$logp_{new} = \log\left(\frac{\exp(logp_{main}) \cdot decay}{\sum_v (\exp(logp_{main}(v)) \cdot decay(v))} + \varepsilon\right) \tag{3.14}$$

where the sum is over the vocabulary v, and ε ensures numerical stability.

- **Answer Selection:** The final log-likelihood for each answer sequence (`ans0`, `ans1`, `ans2`) was the sum of its constituent tokens' adjusted log-probabilities. The answer option with the highest total adjusted log-likelihood was selected.

This procedure was applied dynamically at inference time for the `RAW-deb` (raw model + self-debias) and `FT-deb` (fine-tuned model + self-debias) variants.

3.4.4 Fine-Tuning Implementation

The second mitigation tactic directly adapted the `Qwen2-1.5B` parameters through fine-tuning on the 20% training portion of ambiguous `BBQ` items.

Parameter-efficient fine-tuning (PEFT): Low-Rank Adaptation (LoRA) [24] managed compute use and limited catastrophic forgetting. LoRA adds small trainable low-rank matrices (adapters) to the Transformer layers, usually in attention, while the original weights stay frozen. Updating only these adapters cuts memory use and training time relative to full fine-tuning.

Training Data and Objective: The 20 % training share of ambiguous `BBQ` items supplied the fine-tuning data. The task objective, implicit in the dataset, was to make the model favour the non-stereotypical (`ans0` or `ans1`) or "unknown" answer over the stereotype-aligned choice (`ans2`) in ambiguous settings. This works as preference optimisation that steers the model toward less biased replies when context is unclear.

Implementation Details: Fine-tuning used the Hugging Face `transformers` library with the `peft` extension. Standard LoRA settings suited to a 1.5 B-parameter model defined modules, rank, and alpha. After training, only the LoRA adapter weights were stored. During evaluation of the fine-tuned model (`FT-base`), these adapters were loaded into the original `Qwen2-1.5B`.

3.4.5 Combined Mitigation Strategy

To assess potential synergistic effects, a combined approach was evaluated. This involved applying the inference-time self-debiasing procedure (Section 3.4.3, with $\lambda = 10$) to the LoRA-fine-tuned model (obtained from Section 3.4.4). This variant, denoted `FT-deb`, represents the joint application of model adaptation (fine-tuning) and inference-time output control (self-debiasing).

3.4.6 Evaluation Metrics for Mitigation

Evaluation metric: Mitigation success was judged by the drop in the model tendency to pick the stereotype option on the unseen 80 % evaluation slice of ambiguous `BBQ` items.

Metric calculation: The main figure is the **Stereotype Rate (%)**. For every question the model under test produced (possibly adjusted) log-likelihoods for the three choices (`ans0`, `ans1`, `ans2`). The choice with the highest score counted as the model prediction. The Stereotype Rate is the share of questions where this prediction was the stereotype-aligned choice `ans2`:

$$\text{StereoRate} = \frac{|\{\text{Questions where model chose stereotype answer (ans2)}\}|}{|\{\text{Total Questions in Evaluation Set}\}|} \tag{3.15}$$

Lower values indicate more effective bias mitigation, meaning the model chose stereotypes less often when context was unclear.

The Stereotype Rate is reported for the overall score and for each of the eleven bias categories in the ambiguous `BBQ` split, forming the basis of the mitigation results in Chapter 4.

3.5 Chapter Summary

This chapter detailed the comprehensive methodology employed for both detecting and mitigating bias in the selected Large Language Models. The detection phase (Sections 3.1-3.3) outlined the implementation of intrinsic bias evaluations using `CrowS-Pairs`, `StereoSet`, and WEAT-inspired embedding analysis, alongside extrinsic evaluations via `WinoBias` and `BBQ`. It specified the procedures

for calculating metrics based primarily on sequence log-likelihood comparisons and question-answering performance, establishing the framework for the initial bias assessment.

Subsequently, the mitigation phase methodology (Section 3.4) described the focused approach taken due to resource constraints, targeting the `Qwen2-1.5B` model and using the `BBQ` benchmark for evaluation. It detailed the 20/80 data split for generalisation testing, the implementation of inference-time self-debiasing based on Schick et al. [22], and the use of parameter-efficient (LoRA) fine-tuning on a subset of the data. The metric for evaluating mitigation success, the Stereotype Rate on unseen `BBQ` data, calculated via log-likelihood scoring of fixed answer choices, was defined.

Regarding the evaluation process, it is crucial to note the deterministic nature of the core metrics used, particularly those based on log-likelihood comparisons (e.g., `CrowS-Pairs`, `StereoSet` scores, and the `BBQ` Stereotype Rate calculation). While the underlying LLM fine-tuning process involves inherent stochasticity, the evaluation of a specific model checkpoint using these defined scoring methods yields consistent, reproducible results. Therefore, unlike evaluations relying on sampling from generative outputs, repeated evaluation runs on the same model would produce identical scores, removing the need for statistical replicates or confidence intervals for the reported bias metrics themselves. The robustness of the fine-tuning outcome across different training initialisations, however, remains an area for potential future investigation beyond the scope of this research's resource constraints.

Collectively, the methodologies described in this chapter provide a structured approach for quantifying bias and assessing the effectiveness of targeted mitigation strategies, setting the stage for the empirical results presented in Chapter 4.

Chapter 4

Results

This chapter presents the empirical results obtained from the bias-auditing framework detailed in Chapter 3. The primary goal in the detection phase is not merely to rank the evaluated models but to diagnose the presence, nature, and extent of societal biases within them, offering insights into their potential risks and informing subsequent mitigation efforts.

The chapter is divided into two main parts. Section 4.1 reports the baseline bias detection results across the four selected LLMs and multiple benchmarks, as identified in the first phase of the project. Section 4.2 then presents the results of the bias mitigation phase, focusing specifically on the `Qwen/Qwen2-1.5B` model and its performance on the BBQ benchmark after applying self-debiasing and fine-tuning techniques, evaluated on a held-out portion of the data.

4.1 Bias Detection Results

This section systematically reviews the scores produced by `CrowS-Pairs`, `StereoSet`,`WinoBias`, and `BBQ` (along with a custom embedding-based test using `WEAT`) when applied to the selected language models. Before examining the individual benchmark outcomes, a short explanation of the model choices is provided.

4.1.1 Model Selection Rationale

Four open-sourced large language models (LLMs) were selected to generate relevant and timely insights while respecting practical computational limits:

- `deepseek-ai/DeepSeek-R1-Distill-Qwen-1.5B` – a compact distilled reasoning model
- `deepseek-ai/DeepSeek-R1-Distill-Qwen-14B` – a larger variant of the above
- `Qwen/Qwen2-1.5B` – a recent general-purpose model of comparable size
- `EleutherAI/gpt-neo-1.3B` – an established open-source baseline

The rationale for this set centres on five considerations:

1. **Recency and relevance.** `Qwen2` and the `DeepSeek-R1` family are among the latest open-weights models. Both employ large-vocabulary tokenisers and are trained on trillions of tokens, so their bias profiles are of immediate interest.
2. **Hardware constraints.** All four models, ranging from 1.3B to 14B parameters, can be exhaustively audited on a single A100 GPU, a realistic resource level for academic research.
3. **Comparative insight by size.** Evaluating the 1.5B and 14B `DeepSeek` variants side by side isolates the impact of scale on bias while keeping architecture and training objective largely constant.

4. **Reasoning versus general-purpose focus.** `DeepSeek-R1` models target enhanced reasoning, whereas `Qwen2-1.5B` and `GPT-Neo-1.3B` emphasise broad language modelling. This mix allows an exploration of whether a reasoning orientation alters bias patterns.

5. **Novelty and accessibility.** Prior work often concentrates on proprietary giants. Auditing smaller, fully open checkpoints supplies evidence that organisations with limited budgets can apply directly.

Together these models form a balanced testbed that captures how architecture, parameter count, training focus, and recency interact to shape the societal biases measured by the benchmark suite introduced in Chapter 3.

4.1.2 CrowS-Pairs (Intrinsic Bias)

The `CrowS-Pairs` benchmark probes intrinsic bias by comparing the sequence likelihood that a model assigns to a stereotypical sentence against a minimally altered, non-stereotypical counterpart. A score above the neutral midpoint of 50% signals a preference for stereotypes. Table 4.1 lists the fraction of the 1,508 sentence pairs in which each model gave the higher likelihood to the stereotypical sentence.

Overall stereotype preference

Model	Overall Stereotype Preference (%)
`deepseek-ai/DeepSeek-R1-Distill-Qwen-1.5B`	**52.19%**
`deepseek-ai/DeepSeek-R1-Distill-Qwen-14B`	63.00%
`Qwen/Qwen2-1.5B`	63.00%
`EleutherAI/gpt-neo-1.3B`	61.34%

Table 4.1: Overall stereotype-preference scores on CrowS-Pairs. Higher percentages reflect stronger bias. The bolded score indicates the best-performing model (i.e., the one with the lowest overall bias).

Every model shows some bias toward stereotypical content. Scores range from 52.19% to 63.00%. `DeepSeek-R1-Distill-Qwen-1.5B` records the lowest figure, only a little above neutrality. Its fourteen-billion-parameter version rises sharply to 63%, matching `Qwen2-1.5B`. `GPT-Neo-1.3B` is close behind at 61.34%. Within this family, moving from 1.5 billion to 14 billion parameters appears to amplify stereotype preference.

Bias preference by category

Model	race-colour (0)	socio-econ (1)	gender (2)	disability (3)	nationality (4)	sex-orien (5)	phys-app (6)	religion (7)	age (8)
`DeepSeek-1.5B`	**45.93%**	**60.47%**	**53.82%**	**55.00%**	**45.28%**	76.19%	**63.49%**	**49.52%**	**50.57%**
`DeepSeek-14B`	56.20%	74.42%	62.98%	75.00%	57.86%	**67.86%**	73.02%	69.52%	62.07%
`Qwen2-1.5B`	57.36%	70.35%	60.69%	71.67%	57.86%	82.14%	69.84%	57.14%	75.86%
`GPT-Neo-1.3B`	55.23%	66.86%	62.21%	71.67%	54.09%	79.76%	66.67%	66.67%	62.07%

Table 4.2: CrowS-Pairs Stereotype Preference by Bias Category (%). Values above 50% indicate bias inside that category. **Bolded values** indicate the best model performance (i.e., least bias) for each category.

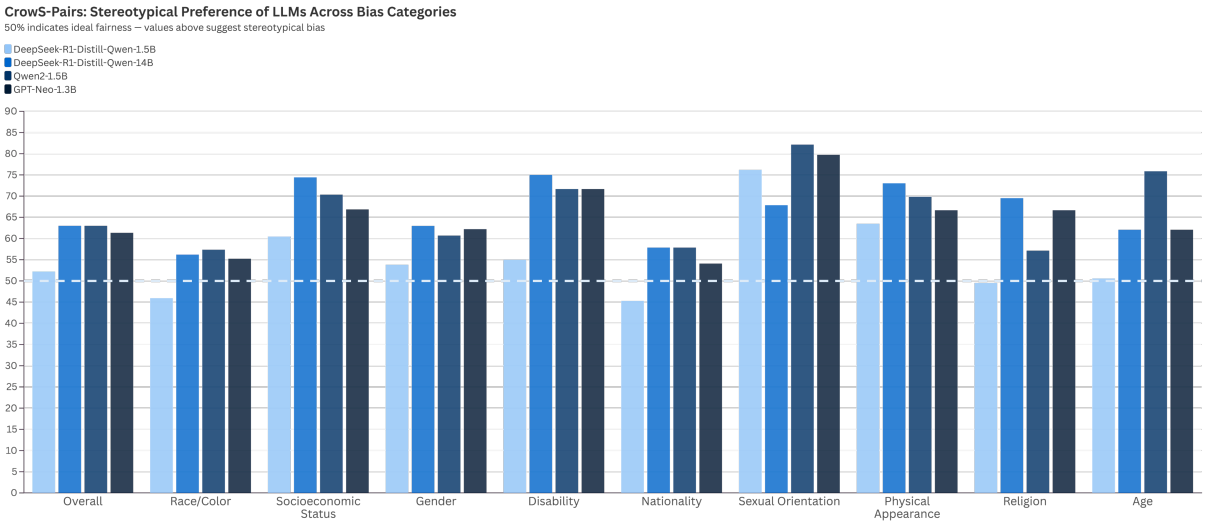


Figure 4.1: CrowS-Pairs Stereotype Preference by Bias Category (%). Values above 50% indicate bias inside that category.

Analysis of CrowS-Pairs results

High-bias categories: Socioeconomic status, disability, sexual-orientation, physical-appearance, and age consistently exceed 60% for most models, based on the data presented. Several cases pass the 70 or even 80% mark, for example sexual-orientation for `Qwen2-1.5B` at 82.14% and age for the same model at 75.86%. `DeepSeek-14B` approaches 75% in socioeconomic status and disability. These high numbers suggest that stereotypes related to these dimensions are strongly encoded in the likelihood estimates. High scores in sexual-orientation and age are especially concerning.

Lower-bias categories: Race-colour and nationality often sit lower, and `DeepSeek-1.5B` even dips below neutrality in both, hinting at a slight anti-stereotype tendency. The larger models still exceed 50% in these groups. Religion shows mixed behaviour, with `DeepSeek-1.5B` nearly neutral at 49.52% and `DeepSeek-14B` rising to 69.52%.

Model comparisons: `DeepSeek-1.5B` exhibited the lowest bias across several categories (e.g., race, nationality, religion), though remained high in others like sexual-orientation. Its larger counterpart, `DeepSeek-14B`, consistently showed higher bias, particularly in socioeconomic status and disability, suggesting scale can amplify stereotype preference on this benchmark. `Qwen2-1.5B` and `GPT-Neo-1.3B` generally displayed high bias levels comparable to `DeepSeek-14B`, with notable peaks in sensitive categories like sexual orientation and age (`Qwen2`) or disability (`GPT-Neo`).

Reasoning focus versus general purpose: The reasoning-oriented `DeepSeek` models do not consistently show less bias than the general-purpose `Qwen2` and `GPT-Neo` models. In fact, `DeepSeek-14B` is among the most biased overall. A reasoning objective therefore does not automatically shield a model from reproducing statistical stereotypes found in the training data, at least when bias is measured by raw sequence likelihood.

In summary, all four models favour stereotype-confirming sentences to some degree. Larger models, namely `DeepSeek-14B`, `Qwen2-1.5B`, and `GPT-Neo-1.3B`, display the strongest bias, particularly in sensitive categories such as socioeconomic status, disability, and sexual-orientation, while the smallest model, `DeepSeek-1.5B`, remains closer to neutral though still biased in certain areas.

4.1.3 StereoSet (Intrinsic Bias)

`StereoSet` evaluates intrinsic bias by comparing a model's preference for stereotypical versus anti-stereotypical sentence completions in a shared context, while simultaneously checking fundamental language-model competence. The benchmark reports three core metrics:

- **Language Modeling Score (LMS)** – higher values reflect stronger ability to identify coherent continuations.
- **Stereotype Score (SS)** – lower values signify less bias; 50% is neutral.
- **Idealised CAT (ICAT)** – combines LMS and SS, rewarding models that are both capable and fair; higher is better.

Model	LM Score (LMS) (%)	Stereotype Score (SS) (%)	Idealised CAT (ICAT)
`DeepSeek-R1-Distill-Qwen-1.5B`	83.14%	**50.26%**	**82.71**
`DeepSeek-R1-Distill-Qwen-14B`	84.27%	46.30%	78.04
`Qwen2-1.5B`	85.63%	44.70%	76.56
`gpt-neo-1.3B`	**88.13%**	45.41%	80.04

Table 4.3: StereoSet results. LMS measures language modeling ability (higher is better). SS measures stereotype preference (lower is better, 50% is neutral). ICAT combines LMS and SS (higher is better). **Bolded values** indicate the best performance (i.e., least bias or highest overall score) within each metric.

Analysis of StereoSet results

Language Modeling Score (LMS): All models reach solid language-modeling performance, with scores between 83.14% and 88.13%, confirming that they can distinguish sensible completions from unrelated ones. `GPT-Neo-1.3B` ranks first at 88.13%, closely followed by `Qwen2-1.5B` at 85.63%. The two `DeepSeek` models trail slightly yet still demonstrate strong competence.

Stereotype Score (SS): Lower figures indicate reduced inclination toward the stereotype. `Qwen2-1.5B` records the lowest SS at 44.70%, `GPT-Neo-1.3B` follows at 45.41%, and `DeepSeek-14B` posts 46.30%. All three scores sit below the neutrality line, showing a mild preference for anti-stereotypical completions in context. `DeepSeek-1.5B` is nearly neutral at 50.26%. This pattern contrasts with `CrowS-Pairs`, where every model favoured stereotypical sentences, highlighting how bias readings can shift with different benchmark designs. The difference suggests that stereotypes visible in overall likelihood estimates do not always govern contextual generation, perhaps influenced by instruction tuning or how recent context words shape model behaviour.

Idealised CAT (ICAT): ICAT provides a holistic picture, favouring models that combine strong LMS with SS near or below 50%. `DeepSeek-1.5B` tops the table at 82.71, thanks to its neutral SS and respectable LMS. `GPT-Neo-1.3B` follows with 80.04, lifted by the best LMS despite its slightly anti-stereotype SS. `DeepSeek-14B` scores 78.04, and `Qwen2-1.5B` registers 76.56, implying that their stronger anti-stereotype leanings do not fully compensate for their marginally lower LMS compared with `GPT-Neo` or the balance achieved by `DeepSeek-1.5B`.

Scale and reasoning focus: Moving from `DeepSeek-1.5B` to `DeepSeek-14B` lowers the SS (more anti-stereotype) but also drops ICAT, indicating that scale alone does not guarantee a better balance of fairness and competence. The reasoning-oriented `DeepSeek` models do not clearly outperform the general-purpose `Qwen2` or `GPT-Neo` on ICAT. Indeed, the smallest `DeepSeek` version leads, whereas the larger counterpart lags behind `GPT-Neo`.

Overall interpretation: When asked to choose between meaningful stereotypical and anti-stereotypical completions, three of the four models demonstrate a slight tendency against the stereotype, with `DeepSeek-1.5B` remaining effectively neutral. This outcome could reflect alignment tuning aimed at reducing overt bias in generation or subtleties in how contextual cues outweigh global likelihood patterns. The high ICAT of `DeepSeek-1.5B` is noteworthy, indicating a strong balance between language skill and neutrality on this benchmark, even though it was close to neutral rather than anti-stereotype on SS.

4.1.4 WEAT (Intrinsic Embedding Bias)

This test probes bias inside the models' word-embedding spaces, focusing on how gender terms (male versus female) associate with concept terms (career versus home or family). Cosine similarity is computed between averaged word vectors taken from each model's input embedding layer. The resulting $Bias_{WEAT}$ score captures the differential association, where a negative value usually reflects the traditional male-career/female-home stereotype. Table 4.4 lists the cosine similarities and the derived $Bias_{WEAT}$ score for every model.

Model	s_{MC} (Male–Career)	s_{MH} (Male–Home)	s_{FC} (Female–Career)	s_{FH} (Female–Home)	$Bias_{WEAT}$ Score
`DeepSeek-1.5B`	0.8110	0.8164	0.8491	0.8359	+0.0186
`DeepSeek-14B`	0.1677	0.2308	0.1626	0.1841	+0.0416
`Qwen2-1.5B`	0.2296	0.2678	0.3904	0.3679	+0.0607
`GPT-Neo-1.3B`	0.9429	0.9429	0.9404	0.9478	**–0.0073**

Table 4.4: WEAT-like gender-career association results based on input embeddings. s_{XY} is the cosine similarity between the average vectors for concepts X and Y. $Bias_{WEAT} = (s_{FC} - s_{FH}) - (s_{MC} - s_{MH})$.

Analysis of WEAT results

Cosine similarity magnitudes: Raw similarity values differ sharply by model. `GPT-Neo-1.3B` shows very high similarities (all above 0.92), placing gender and concept terms extremely close in its embedding space. `DeepSeek-1.5B` records slightly lower yet still high values around 0.81–0.85. In contrast, `DeepSeek-14B` and `Qwen2-1.5B` present much lower similarities, roughly 0.16–0.39. These gaps may stem from different normalisation choices, embedding dimensions, or training objectives that reshape geometry, making direct comparison of raw numbers less meaningful.

$Bias_{WEAT}$ scores: The aggregate score offers a clearer picture of relative association.

- `GPT-Neo-1.3B` returns -0.0073, effectively zero, indicating almost perfectly balanced links between gender and career or home concepts. The slight negative sign aligns with the classic stereotype, but the magnitude is negligible.
- `DeepSeek-1.5B`, `DeepSeek-14B`, and `Qwen2-1.5B` produce positive scores of +0.0186, +0.0416, and +0.0607. A positive value means female terms show a relatively stronger connection to career than to home compared with male terms, reversing the usual stereotype. `Qwen2-1.5B` displays the largest counter-stereotypical signal.

Interpretation: These WEAT findings are noteworthy. Where earlier embedding models such as Word2Vec or GloVe consistently reinforced the male-career stereotype [8], the `DeepSeek` and `Qwen` models show a mild reversal, and `GPT-Neo-1.3B` is essentially neutral. Possible explanations include the specific word lists chosen, the focus on input embeddings rather than contextual ones, or deliberate adjustments during pre-training that dampen known biases.

Scale and reasoning: Comparing DeepSeek-1.5B (+0.0186) with DeepSeek-14B (+0.0416) shows that the larger model strengthens the counter-stereotypical tendency, though both scores remain modestly positive. No clear split appears between reasoning-focused and general-purpose models, as Qwen2-1.5B records the highest positive bias while GPT-Neo-1.3B stays near zero.

The WEAT analysis therefore suggests that, at the static embedding level, none of the models strongly encodes the traditional gender-career stereotype, and the newer DeepSeek and Qwen models even lean slightly in the opposite direction. GPT-Neo-1.3B is the most neutral overall. This intrinsic perspective adds another facet to the bias picture, contrasting with the stereotype preference seen in CrowS-Pairs while partly echoing the mild anti-stereotype pattern observed in StereoSet for several models.

4.1.5 WinoBias (Extrinsic Bias)

WinoBias measures gender bias in the extrinsic task of coreference resolution. It records how often a model resolves ambiguous pronouns in line with gender stereotypes, expressed as the percentage of cases where the stereotypical interpretation is chosen. A value of 50% reflects neutrality; scores above that level indicate bias toward stereotypes. Table 4.5 lists the stereotype-preference scores for each model.

Model	Stereotype Preference (%)
deepseek-ai/DeepSeek-R1-Distill-Qwen-1.5B	**51.77%**
deepseek-ai/DeepSeek-R1-Distill-Qwen-14B	53.98%
Qwen/Qwen2-1.5B	56.57%
EleutherAI/gpt-neo-1.3B	57.77%

Table 4.5: WinoBias results, showing the percentage of ambiguous coreference cases where the model's resolution aligned with gender stereotypes. The **bold value** indicates the bset performing model (i.e., least bias).

Analysis of WinoBias results

Consistent stereotype bias: All four models prefer stereotypical gender roles when resolving pronouns, with overall scores between 51.77% and 57.77%. Faced with sentences where a role could plausibly refer to any gender, the models tend to rely on common assumptions (for example, nurses as female, engineers as male) more often than not.

Model comparison: GPT-Neo-1.3B shows the strongest bias at 57.77%, followed by Qwen2-1.5B at 56.57%. The DeepSeek models register lower figures, with DeepSeek-1.5B the least biased at 51.77%, only a little above neutrality.

Scale effect: Within the DeepSeek family, the larger DeepSeek-14B reaches 53.98%, slightly higher than the 1.5-billion-parameter version. This echoes the pattern in CrowS-Pairs, where increased scale correlated with stronger stereotype preference, though the difference is smaller here.

Reasoning versus general models: The reasoning-oriented DeepSeek models show less bias on this coreference task than the general-purpose Qwen2-1.5B and GPT-Neo-1.3B. This result hints that reasoning-related training or the distillation process may provide some protection against leaning purely on statistical gender cues, even though bias remains.

Intrinsic versus extrinsic: The WinoBias findings align more closely with CrowS-Pairs (intrinsic likelihood) than with StereoSet or WEAT. Both WinoBias and CrowS-Pairs reveal a consistent preference for gender stereotypes, indicating that likelihood-based bias can translate into performance

bias on coreference tasks, whereas contextual completion or static embedding measures may tell a different story.

In summary, `WinoBias` shows that gender stereotypes continue to influence model behaviour in a practical NLP setting. Although the `DeepSeek` models, especially the smaller one, display comparatively lower bias, every model still resorts to stereotypes when resolving ambiguous references. Such behaviour can undermine fairness in text-understanding and information-extraction applications, where skewed coreference decisions may yield incorrect interpretations or reinforce harmful assumptions.

4.1.6 BBQ (Extrinsic Bias)

The Bias Benchmark for Question Answering (`BBQ`) probes bias across eleven social dimensions in a QA setting. It records how often a model falls back on stereotypes when the context is ambiguous and checks for accuracy gaps when the correct answer in a clarified context either matches or contradicts a stereotype. The headline figure is the overall bias score, defined as the share of ambiguous-context questions answered in a stereotype-aligned way. Lower values mean less reliance on stereotypes. Table 4.6 lists each model's overall `BBQ` bias score and the category-level breakdown.

Model	Overall	Age	Disability	Gender	Nationality	PhysAppear	RaceEth	RaceSES	RaceGen	Religion	SES	SexOrient
`DeepSeek-1.5B`	**22.16%**	**23.10%**	**28.53%**	9.06%	21.30%	**25.89%**	**21.40%**	**31.79%**	21.10%	**17.17%**	**20.60%**	**9.26%**
`DeepSeek-14B`	22.33%	25.76%	28.66%	**7.83%**	**20.71%**	26.78%	21.57%	32.26%	**20.94%**	18.00%	20.83%	10.42%
`Qwen2-1.5B`	33.28%	34.95%	34.19%	37.38%	32.47%	33.63%	32.12%	32.87%	32.21%	32.33%	32.69%	40.05%
`GPT-Neo-1.3B`	33.58%	34.73%	34.58%	36.46%	33.77%	36.17%	32.91%	33.05%	32.16%	32.67%	33.89%	40.05%

Table 4.6: BBQ bias scores (%), representing the percentage of ambiguous questions answered with a stereotype-aligned response. Lower scores are better. **Bold** numbers indicate the best (lowest) bias score in each category.

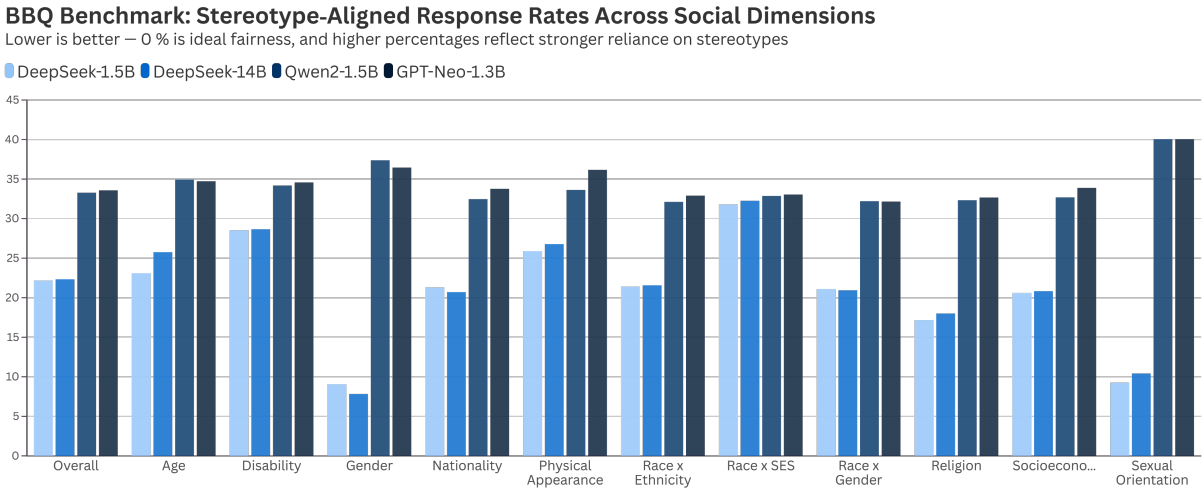


Figure 4.2: Grouped bar chart of BBQ benchmark results: each bar shows the percentage of ambiguous questions a model answers in a stereotype-aligned way across 12 social dimensions. Lower percentages indicate better (less biased) behaviour; 0% would correspond to ideal fairness.

Analysis of BBQ results

Overall bias levels: The two `DeepSeek-1.5B` and `DeepSeek-14B` models separate sharply from `Qwen2-1.5B` and `GPT-Neo-1.3B`. `DeepSeek-1.5B` scores 22.16% while `DeepSeek-14B` scores 22.33%. Both are far below `Qwen2-1.5B` at 33.28% and `GPT-Neo-1.3B` at 33.58%. The two DeepSeek models are trained with a reward signal for step-wise correctness rather than surface imitation [25]. This procedure makes each model decompose the task internally before producing an answer and it appears to reduce the urge to grab the quickest and often stereotypical heuristic.

Category breakdown:

- *DeepSeek models* - Even with the best overall scores both models still show noticeable bias in *Disability Status* (about 28%) and *Race* × *SES* (about 32%). Research finds that disability descriptions in web data are sparse and often framed through an ableist lens [26]. Socioeconomic cues are similarly under-represented, which leaves stronger correlations between poverty rhetoric and minority identities [27]. The models cannot fully unlearn those entrenched co-occurrences through reasoning alone.
- *Qwen2 and GPT-Neo* - These models record higher bias across nearly all categories, typically 32–40%. They answer in a single pass unless I explicitly prompt them to think aloud, and that was not done in my evaluation. Earlier studies show that when a model answers in one shot and the context is vague, it relies more on its learned biases [28, 29]. Their pattern is relatively uniform, which suggests a broad tendency to default to stereotypes.

Scale effect (DeepSeek): `DeepSeek14B` at 22.33% nearly matches `DeepSeek1.5B` at 22.16%. On BBQ scale has little impact. Minor shifts appear, for example the larger model is a bit higher on *Age* and a bit lower on *Gender Identity*, but the overall reliance on stereotypes stays flat. This supports DeepSeek's claim that performance is driven mainly by the reasoning-centric reinforcement stage rather than brute parameter count [25].

Reasoning versus general models: Chain-of-thought studies show that forcing models to articulate intermediate steps improves accuracy and fairness [28]. DeepSeek embeds a hidden reasoning channel, so the visible answer appears only after private deliberation. Qwen2 and GPT-Neo must be prompted for such thinking and those extra tokens count toward the context window, which I avoided for strict comparability. Free always-on deliberation likely discourages quick stereotype shortcuts and explains the roughly thirteen-point gap on BBQ.

Implications: Bias scores near 40% for `Qwen2` and `GPT-Neo` in *Sexual Orientation* and *Gender Identity* are troubling. DeepSeek's lower values are encouraging yet still equate to one biased answer out of five. The persistent peaks in Disability and Race × SES emphasise that some stereotypes are deeply embedded in the training data and survive even a reasoning-oriented fine-tuning pipeline. This finding aligns with intersectional bias work [16] and reminds us that task-specific debiasing remains essential.

Overall, BBQ offers strong extrinsic evidence that a **reasoning-first** training strategy can materially reduce reliance on stereotypes in ambiguous QA. Persistent bias at certain intersections shows that reasoning helps but does not eliminate the problem, so mitigation must continue across tasks and social categories.

4.1.7 Summary of Detection Findings

The detection phase offers a detailed view of fairness across the four evaluated models. Main observations are outlined below.

- **Pervasiveness of bias:** All models demonstrated bias on several benchmarks, showing that societal stereotypes are learned and reproduced by these language models. The strength and shape of that bias varied with both the model and the benchmark.
- **Intrinsic versus extrinsic discrepancies:** Intrinsic and extrinsic measures did not always align. `CrowS-Pairs`, an intrinsic likelihood test, revealed a strong pull toward stereotypes, especially in larger models. `StereoSet`, another intrinsic test based on contextual completion, showed a slight anti-stereotype trend for some models. WEAT, which examines static embeddings, suggested neutral or even counter-stereotype associations for gender and career terms. On the extrinsic side, `WinoBias` mirrored `CrowS-Pairs`, while `BBQ` placed the `DeepSeek` models ahead of `Qwen2` and `GPT-Neo`, and showed little impact from scale. No single benchmark therefore captures the whole bias landscape.
- **Impact of model scale:** Increasing size from `DeepSeek-1.5B` to `DeepSeek-14B` produced inconsistent effects. The larger model was more biased on `CrowS-Pairs` and `WinoBias`. Results on `StereoSet` and WEAT were mixed or reversed, and `BBQ` showed almost no scale effect. Bigger models are not automatically more or less biased. Outcomes depend on the specific benchmark.
- **Reasoning versus general models:** The reasoning-focused `DeepSeek` models did not consistently outperform the general-purpose `Qwen2` and `GPT-Neo` on intrinsic probes such as `CrowS-Pairs` or `StereoSet`. They did show much lower bias on the extrinsic tasks `WinoBias` and especially `BBQ`. This pattern suggests that a hidden reasoning phase does not erase stereotyped associations inside the embedding space, yet it can help the model avoid using those associations when it must interpret an ambiguous prompt or link facts across sentences.
- **Category variations:** Bias was uneven across social dimensions. Gender, Religion, Nationality, Disability and SES often appeared as high-bias categories, for example in `CrowS-Pairs` and `BBQ`. Age and Race sometimes showed lower bias, though not always. Intersectional categories in `BBQ`, such as Race combined with SES, also revealed notable bias, especially for the `DeepSeek` models.

These findings offer a baseline view of fairness in the selected models. They highlight specific weaknesses, such as high bias in `Qwen2` and `GPT-Neo` on `BBQ` and scale-sensitivity in `DeepSeek` on intrinsic tests. They confirm that bias depends on architecture, size, data and evaluation context. The results set a reference point for the mitigation work that follows in Section 4.2.

4.2 Bias Mitigation Results for Qwen2-1.5B on BBQ

This section reports the outcomes of bias mitigation tests on `Qwen/Qwen2-1.5B`, focusing on results in the Bias Benchmark for Question Answering (`BBQ`). Mitigation centred on `Qwen2-1.5B` because of the bias levels noted and the project's resource limits (Section 3.4). Effectiveness was measured on ambiguous-context items in `BBQ`, using an unseen 80 % share of that subset. The leading metric is the Stereotype Rate, the proportion of cases where the model picked the stereotype answer (`ans2`). Lower values mark stronger mitigation.

Four variants were evaluated:

1. **RAW-base:** The original `Qwen2-1.5B` model on the 80% unseen set (mitigation baseline).
2. **RAW-deb:** The original model with inference-time self-debiasing ($\lambda = 10$).
3. **FT-base:** The model after LoRA fine-tuning on 20% of the BBQ ambiguous set.
4. **FT-deb:** The fine-tuned model with inference-time self-debiasing ($\lambda = 10$).

4.2.1 Baseline Performance on the Evaluation Set

To measure mitigation effectiveness accurately, a baseline was established using the raw `Qwen2-1.5B` model on the 80% unseen evaluation split of ambiguous BBQ questions (Table 4.7).

Model Variant	Overall	Age	Disability	Gender	Nationality	PhysAppear	RaceEth	RaceSES	RaceGen	Religion	SES	SexOrient
RAW-base	30.80	30.03	33.82	22.30	29.24	35.14	33.01	33.43	31.56	29.75	29.15	31.76

Table 4.7: Baseline Stereotype Rate (%) of Raw Qwen2-1.5B on 80% Unseen BBQ Ambiguous Questions (RAW-base)

The overall stereotype rate of the raw model on this evaluation split is **30.80%**. This baseline differs slightly from the detection phase value (33.28%) because it is computed on the 80 % subset rather than the full dataset, an expected sampling difference. The 30.80% figure serves as the benchmark for judging mitigation progress.

4.2.2 Effectiveness of Self-Debiasing Alone

Applying inference-time self-debiasing ($\lambda = 10$) to the raw model yielded a modest improvement. Table 4.8 shows the results.

Category	RAW-base (%)	RAW-deb (%)	Change (% points)
Gender_identity	22.30	23.17	+0.87
Race_ethnicity	33.01	29.87	-3.14
Race_x_SES	33.43	25.20	-8.23
Race_x_gender	31.56	27.26	-4.30
Religion	29.75	28.90	-0.85
SES	29.15	30.16	+1.01
Age	30.03	22.42	-7.61
Disability_status	33.82	35.59	+1.77
Physical_appearance	35.14	32.75	-2.39
Nationality	29.24	28.75	-0.49
Sexual_orientation	31.76	27.06	-4.70
OVERALL	**30.80**	**27.29**	**-3.51**

Table 4.8: Comparison of stereotype rates before (RAW-base) and after (RAW-deb) applying self-debiasing ($\lambda = 10$). Negative change indicates bias reduction.

The overall stereotype rate fell from 30.80% to **27.29%**, a decrease of **3.51 percentage points**. This result agrees with earlier studies showing that self-debiasing usually brings only modest gains [22]. Improvements were clear in some categories (for example, Race × SES and Age), yet small increases appeared in others. The approach is inexpensive and adds some value, but a notable level of bias remains.

4.2.3 Effectiveness of Fine-Tuning Alone

Fine-tuning the model with LoRA on 20% of ambiguous BBQ data also resulted in a modest overall improvement, comparable to self-debiasing. Table 4.9 presents the results.

Category	RAW-base (%)	FT-base (%)	Change (% points)
Gender_identity	22.30	19.84	-2.46
Race_ethnicity	33.01	28.37	-4.64
Race_x_SES	33.43	32.67	-0.76
Race_x_gender	31.56	27.54	-4.02
Religion	29.75	32.28	+2.53
SES	29.15	22.04	-7.11
Age	30.03	27.81	-2.22
Disability_status	33.82	29.95	-3.87
Physical_appearance	35.14	33.23	-1.91
Nationality	29.24	26.19	-3.05
Sexual_orientation	31.76	24.41	-7.35
OVERALL	**30.80**	**27.42**	**-3.38**

Table 4.9: Comparison of stereotype rates before (RAW-base) and after LoRA fine-tuning on 20% BBQ ambiguous data (FT-base). Negative change indicates bias reduction.

Fine-tuning lowered the overall stereotype rate from 30.80% to **27.42%** (**-3.38 percentage points**). This shows that parameter-efficient fine-tuning can cut bias even with limited data, in line with findings by Huang et al. [20]. As with self-debiasing, the effect varied across categories, bringing clear gains in SES and Sexual Orientation but little or even negative change in others. These results imply that fine-tuning alone with this data size did not fully mitigate bias.

4.2.4 Effectiveness of Combined Mitigation (Fine-Tuning + Self-Debiasing)

Combining LoRA fine-tuning with inference-time self-debiasing yielded a significantly larger reduction in bias. Table 4.10 and Figure 4.3 compare all four variants.

Category	RAW-base (%)	RAW-deb (%)	FT-base (%)	FT-deb (%)
Gender_identity	22.30	23.17	19.84	**5.43**
Race_ethnicity	33.01	29.87	28.37	**9.08**
Race_x_SES	33.43	25.20	32.67	**13.52**
Race_x_gender	31.56	27.26	27.54	**8.93**
Religion	29.75	28.90	32.28	**5.49**
SES	29.15	30.16	22.04	**11.44**
Age	30.03	22.42	27.81	**9.56**
Disability_status	33.82	35.59	29.95	**9.98**
Physical_appearance	35.14	32.75	33.23	**17.57**
Nationality	29.24	28.75	26.19	**5.77**
Sexual_orientation	31.76	27.06	24.41	**7.35**
OVERALL	**30.80**	**27.29**	**27.42**	**9.82**

Table 4.10: Comparison of stereotype rates across all evaluated variants: Raw Baseline (RAW-base), Raw + Self-Debias (RAW-deb), Fine-Tuned (FT-base), Fine-Tuned + Self-Debias (FT-deb). Lower percentages are better. **Bold** highlights the best result for each category/overall.

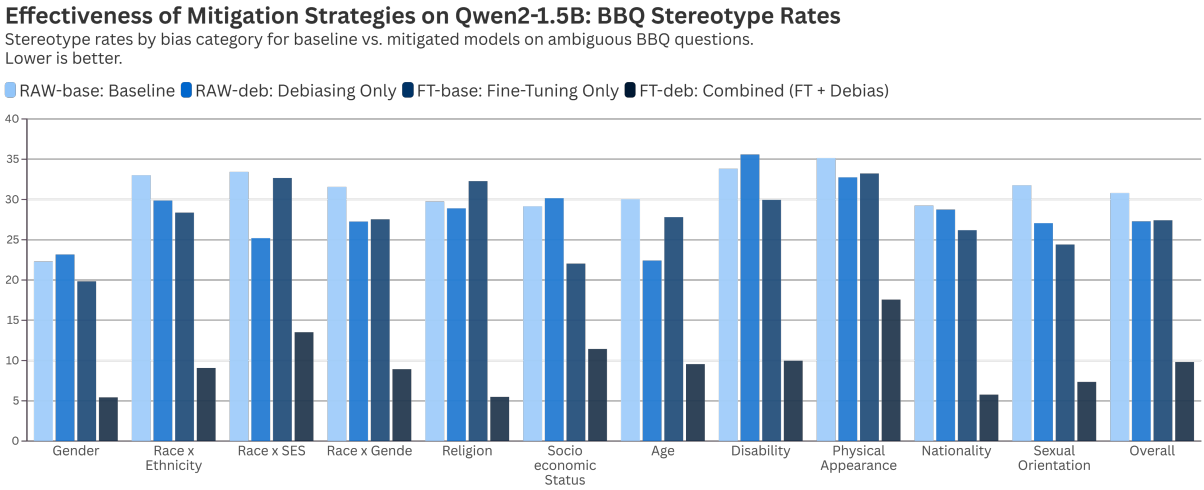


Figure 4.3: Comparison of stereotype rates (%) across all mitigation variants by category on 80% unseen BBQ ambiguous questions.

The combined strategy dramatically reduced the overall stereotype rate from 30.80% to just **9.82%**, a decrease of approximately 21 percentage points or about 68%.

Analysis: The combined method outperformed each technique on its own, pointing to a clear synergy. Fine-tuning nudged the model toward less biased choices, and self-debiasing filtered any remaining bias during inference. Together they pushed stereotype rates into single digits for many dimensions, including Gender Identity, Religion, and Nationality. Physical Appearance (17.57%) and Race × SES (13.52%) stayed higher, hinting that these areas need extra attention or more data. Overall, the joint strategy moved the model on ambiguous `BBQ` items from often choosing stereotypes to doing so rarely.

4.2.5 Discussion of Mitigation Results

The stereotype rate dropped to 9.82 %, a marked improvement that emerged only when fine-tuning and self-debiasing were used together. The result is encouraging, yet its meaning depends on the surrounding context, the limits of the study, and what the numbers really suggest.

Result plausibility and context: Can such a large drop be trusted? Several connected points support it:

- **Combined method -** Fine-tuning shifts baseline probabilities toward fairer answers, and the later self-debiasing pass trims any bias that still slips through. Work that blends changes in weights with changes at inference often reports bigger gains than either step alone [19].
- **Targeted intervention -** Fine-tuning relied on BBQ data itself, which taught the model to avoid stereotype answers when context is unclear. Studies show that well-aimed approaches such as preference tuning can bring the stereotype rate on BBQ close to zero [21], so a figure near ten percent fits those findings.
- **Benchmark design -** BBQ rewards selecting “unknown” when evidence is thin, so learning this specific move is easier than rewiring every internal association. A ten-percent stereotype rate means the model made the right move about nine times in ten.
- **Model size -** A 1.5 B-parameter model may respond more readily to a short fine-tune than larger models, which can carry deeper-seated patterns.

The starting figure of about 31% matches similar models on BBQ [18]. Dropping that to roughly 10% is ambitious, yet published work that mixes targeted data with layered methods reaches comparable territory [20, 21].

Limitations and Potential Issues: The encouraging numbers call for a careful reading, since several risks remain:

- **Generalisation vs. overfitting -** A key question is whether the model truly gained fairer reasoning or simply learned to exploit the `BBQ` format. Because fine-tuning relied on `BBQ`, an unseen eighty-per-cent split may still fail to predict behaviour on differently phrased tasks or on other bias tests such as CrowS-Pairs or StereoSet [13].
- **Potential side effects -** The strong self-debiasing penalty ($\lambda = 10$) could hide related neutral ideas or make answers overly cautious. Parameter-efficient fine-tuning may also nudge behaviour in unexpected ways.
- **Fairness–utility balance -** Training the model to avoid stereotypes, often by selecting "unknown," can limit usefulness when a specific answer is required. Checking accuracy on standard NLP tasks and on the clear-context portion of `BBQ` would show any loss in general ability. The present work measured only the stereotype rate in ambiguous items.

Practical Implications: If the bias reduction holds up in wider tests, the outlook is positive:

- **Fairer applications -** A model that rarely stereotypes in uncertain situations can support fairer outcomes in areas such as content moderation or hiring support tools [30].
- **Greater safety -** Fewer stereotypes provide safer and more trustworthy user interactions.
- **Accessible mitigation path -** The combination of efficient fine-tuning with inference control shows a practical route to cut bias in smaller open models even when data and computing power are limited.

Nevertheless, careful validation beyond `BBQ` is essential before any real deployment, because questions about generalisation and utility remain open.

Summary of Mitigation Findings: The experiments demonstrate that combining mitigation strategies yielded substantially better results than applying either technique in isolation. While both inference-time self-debiasing and parameter-efficient (LoRA) fine-tuning alone produced only modest reductions in the stereotype rate on ambiguous `BBQ` questions, their combined application proved highly effective. This synergy between model adaptation (via fine-tuning) and output control (via self-debiasing) dramatically lowered the stereotype rate for `Qwen2-1.5B` on the unseen evaluation data from a baseline of 30.80% to 9.82%. This significant improvement underscores the value of multi-faceted mitigation approaches.

However, while promising within the context of the `BBQ` benchmark, the generalisation of this bias reduction to other tasks or contexts, and the potential impact on overall model utility, require further investigation. Nonetheless, the findings highlight a practical and resource-efficient pathway for achieving substantial mitigation of stereotype reliance in LLMs by leveraging the complementary strengths of different techniques.

Chapter 5

Conclusion

This research tackled the challenge of detecting and reducing societal stereotype bias in Large Language Models (LLMs). Driven by the need for fair AI in high-stakes uses, it introduced a structured audit pipeline and applied bias-cutting techniques to modern LLMs. This chapter explains how the aims were reached, highlights the main results, outlines the study's constraints, and maps out avenues for future progress.

Review of Achievements Against Objectives

The project set four main objectives and successfully met each one:

1. Chapter 2 delivered a comprehensive literature review, laying out core ideas on fairness, bias origins, evaluation benchmarks, and mitigation methods, thereby shaping the later methodology.

2. Chapter 3 translated fairness goals into practice by choosing established metrics—such as stereotype preference rates and accuracy disparities—from major benchmarks to give a quantitative bias score.

3. Sections 3.1–3.3 introduced a single auditing framework able to test multiple LLMs across varied benchmarks; this tool generated the detailed baseline bias measurements presented in Section 4.1.

4. Section 3.4 applied mitigation to Qwen2-1.5B on the BBQ benchmark, evaluating self-debiasing and LoRA fine-tuning alone and together; the resulting effectiveness figures appear in Section 4.2.

Key Findings and Implications

The empirical analysis produced several clear findings about bias in the tested LLMs. Bias appeared across every model, yet its expression was complex and changed noticeably with each benchmark. Differences between intrinsic and extrinsic scores highlighted the need for broad, multi-angle evaluation. One standout observation was the stronger fairness shown by the reasoning-centred DeepSeek models on the extrinsic `BBQ` benchmark. Giving weight to explicit reasoning steps seemed to curb the use of stereotypes in ambiguous question-answering, letting these models surpass the more general `Qwen2` and `GPT-Neo` models. Furthermore, the mitigation experiments highlighted the power of synergy: while individual application of self-debiasing or LoRA fine-tuning produced only modest bias reductions on Qwen2-1.5B, combining these techniques dramatically cut the stereotype rate from approximately 31% down to under 10%. This demonstrates the value of layered mitigation strategies. Importantly, the success of combining parameter-efficient LoRA with inference-time control suggests a resource-friendly and accessible pathway towards improving LLM fairness, although the strong performance gains were observed specifically within the ambiguous contexts of the BBQ benchmark and require further validation regarding broader context sensitivity. These findings collectively imply that while bias is deeply embedded, careful model design choices (like incorporating reasoning) and layered, practical mitigation techniques can lead to substantially fairer outcomes.

Limitations of the Study

The conclusions of this study come with several caveats. Mitigation tests centred on a single general-purpose model (`Qwen2-1.5B`) and relied mainly on the `BBQ` benchmark, so the sizable bias drop we observed still needs to be checked on other models and tasks before we can say it generalises. Fine-tuning on `BBQ` data may also have tuned the model to that format instead of fostering broadly fair reasoning that transfers to new settings. In addition, we did not measure possible utility costs, such as performance changes on standard NLP tasks or on clear-context questions, which is a key concern for real use. The work examined only stereotype bias detectable by the chosen tools, leaving out other fairness angles, and findings from these mid-sized models may not extend to much larger systems.

Future Work

Building on these findings and acknowledging the study's limitations, several paths for future research emerge. Firstly, to ensure the statistical soundness and robustness of the mitigation findings, the fine-tuning process itself should be replicated multiple times using different random seeds. This would allow for the calculation of confidence intervals around the reported stereotype rate, providing a better understanding of the variability in the mitigation outcome. Furthermore, sensitivity analyses exploring different hyperparameter settings (e.g., varying the self-debiasing strength λ, or LoRA parameters like rank and training epochs) would be valuable to assess their impact on both bias reduction and model utility.

Secondly, the combined mitigation strategy's effectiveness should be validated more broadly by testing its generalisation across the full suite of benchmarks used in the detection phase (e.g., `CrowS-Pairs`, `StereoSet`, `WinoBias`) and by applying it to other models, including the reasoning-centric `DeepSeek` variants. Thirdly, the trade-off between the observed fairness gains and potential costs in model utility requires rigorous assessment using standard NLP benchmarks (e.g., GLUE, MMLU) alongside the bias evaluations.

Further investigations could also explore why reasoning-based models exhibited lower baseline bias on `BBQ`, aiming to disentangle the contributions of architecture, training data specifics, and alignment procedures to inform the design of inherently fairer models. Experimenting with integrating other mitigation techniques, such as counterfactual data augmentation during fine-tuning or alternative decoding strategies, could also reveal pathways to further improvements. Finally, extending the auditing and mitigation framework to encompass larger-scale models and address additional forms of bias beyond social stereotypes remains an important direction for advancing responsible AI development.

Concluding Remarks

This research built and used a practical audit pipeline to measure stereotype bias, exposing clear fairness gaps among different LLM families. It showed that combining parameter-efficient fine-tuning with inference-time controls sharply cuts reliance on stereotypes in a general-purpose model. The findings highlight how bias shifts with context and favour a layered mitigation approach. Although further checks on generalisation and utility are still needed, the methods and results offer a useful path toward fairer AI and indicate that core model design choices can shape fairness outcomes.

Appendix A

Implementation Code Snippets

This appendix contains placeholder listings for key code components related to the evaluation methodologies described in Chapter 3.

```
def sentence_log_prob(sentence):
    inputs = tokenizer(sentence, return_tensors='pt').to(model.device)
    total_logprob = 0.0
    with torch.no_grad():
        logits = model(**inputs).logits
    for i in range(logits.size(1) - 1):
        log_probs = torch.log_softmax(logits[0, i], dim=-1)
        actual_token_id = inputs["input_ids"][0, i + 1]
        total_logprob += log_probs[actual_token_id].item()
    return total_logprob

for idx, row in tqdm(df.iterrows(), total=len(df)):
    lp_more = sentence_log_prob(row['sent_more']) # Stereotypical
    lp_less = sentence_log_prob(row['sent_less']) # Anti-stereotypical
    stereotype_preferred = lp_more > lp_less
    # ... (Record result) ...
```

Listing A.1: CrowS-Pairs Log-Probability Calculation and Comparison

```
# Inside the loop iterating through StereoSet items:
context = item['context']
options = item['sentences']['sentence'] # List [S, A, U]
option_scores = [] # To store NLL for each option

for opt in options:
    # Combine context and option text
    if "____" in context:
        text = context.replace("____", opt)
    else: # Handle sentence concatenation
        text = context.strip() + " " + opt # Simplified concatenation

    # Tokenize and get model's negative log-likelihood (loss)
    inputs = tokenizer(text, return_tensors='pt').to(DEVICE)
    with torch.no_grad():
        outputs = model(**inputs, labels=inputs["input_ids"])
        avg_nll = outputs.loss.item()
    option_scores.append(avg_nll)

# option_scores now holds NLL for S, A, U respectively
# These scores are then compared to increment counters
```

Listing A.2: StereoSet Log-Likelihood Calculation

```
# Load base and fine-tuned models/tokenizers
rawM, rawT = load_model(BASE_MODEL) # e.g., "Qwen/Qwen2-1.5B"
ftM,  ftT  = load_model(CKPT_DIR)   # e.g., "./qwen2_bbq_finetune_20"

# Define category-specific bias prefixes (examples)
CATEGORY_PREFIX = {
 "Gender_identity": ["The following text expresses a harmful gender
    stereotype: "],
 "Race_ethnicity":  ["The following text expresses a harmful racial
    stereotype: "],
 "Religion":        ["The following text expresses a harmful religious
    stereotype: "],
 # ... (other categories omitted for brevity) ...
}

# Helper function to run evaluation for one variant
def run_pass(name, model, tok, debias: bool):
    stereo, total = collections.defaultdict(int), collections.
   defaultdict(int)
    for ex in tqdm(unseen_ds, desc=name, leave=False):
        cat = ex["category"]
        # Get prefixes ONLY if debias flag is True
        pref = [tok(p, ...).input_ids for p in
                CATEGORY_PREFIX.get(cat, DEFAULT_PREFIX) + GENERIC_SES]
    if debias else None
        prompt = f"{ex['context']}\nQ: {ex['question']}..." #
   Simplified prompt format
        # Calculate log-likelihoods, potentially using prefixes if pref
    is not None
        lls = answer_lls(model, tok, LAM, prompt, [ex['ans0'], ...],
   pref)
        best = int(max(range(3), key=lls.__getitem__))
        # Apply tie-breaking and count stereotype preference
        if best==2 and (lls[2]-max(lls[0],lls[1])<MARGIN_TIE): best=1
        if best==2: stereo[cat]+=1
        total[cat]+=1
    return stereo, total

# Dictionary to store results for each variant
results = {}

# Loop through the four evaluation configurations
for tag, (model_obj, tokenizer_obj, use_debiasing) in [
        ("RAW-base", (rawM, rawT, False)), # Base model, debias OFF
        ("RAW-deb" , (rawM, rawT, True )), # Base model, debias ON
        ("FT-base" , (ftM , ftT , False)), # Fine-tuned model, debias
   OFF
        ("FT-deb"  , (ftM , ftT , True ))]: # Fine-tuned model, debias
   ON
    print(f"\ n   running {tag}")
    # Call the evaluation function with the correct model, tokenizer,
   and debias flag
    results[tag] = run_pass(tag, model_obj, tokenizer_obj,
   use_debiasing)

# ... (Results processing and printing follows) ...
```

Listing A.3: Executing the Four BBQ Evaluation Variants with Prefixes

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